



Parametric Analysis and Machine Learning Model for Compressive Strength of Rice Husk Ash Concrete

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Abstract

Growing demand for sustainable construction materials has increased interest in using Rice Husk Ash (RHA) as a supplementary cementitious material to enhance concrete performance. However, traditional testing is costly and labor-intensive. This study proposes a predictive model using a Hybrid Neuro-Genetic Algorithm (ANN-GA) to efficiently estimate the compressive strength of RHA concrete. A comprehensive database of 1,192 datasets was utilized, encompassing seven key parameters: water, cement, RHA, fine aggregates, coarse aggregates, superplasticizer, and curing age. The ANN-GA model, configured with a single hidden layer of 36 neurons, demonstrated the best performance metrics. Comparative analysis against existing models showed the ANN-GA's superiority, yielding the highest R value of 0.988, the lowest MAE of the 3.065, lowest RMSE of 3.883, and the least PE of 2.772%. Additionally, a parametric analysis was conducted to identify the effect of each input parameter. The findings confirm the capability of the ANN-GA model as a reliable and efficient alternative to traditional methods for predicting the compressive strength of RHA concrete, while also providing insights into optimal mix design strategies for enhanced performance.

1 Introduction

The construction industry seeks to explore innovative materials and optimize the performance of the most critical construction material through advanced techniques. With the immense production of ordinary portland cement (OPC), environmental concerns regarding the high carbon footprint output intensified [1]. Rice husk ash (RHA) has established a sustainable partial replacement for cement to oppose the increasing demand for OPC [2]. The compressive strength of concrete is a fundamental mechanical property for assessing its reliability and quality. Several studies have shown that compressive strength of concrete with RHA replacement typically varies based on its mix design, curing process, and intended application. Because rice husk ash can produce concrete with compressive strengths that are comparable to traditional pure cement mixes—typically between 10 and 45 MPa—RHA has drawn a lot of interest as a supplemental

cementitious material. Furthermore, RHA makes it possible to produce high-strength concrete (HSC) like Grade 60 and Grade 80 in an economical and effective manner, frequently outperforming control concrete mixes [3–5]. In addition to improving mechanical qualities, RHA provides a sustainable solution by reducing the environmental impact of the production of regular Portland cement, which contributes to 8–10% of global CO₂ emissions. Interest in using agricultural and industrial wastes, such as RHA, as environmentally friendly pozzolanic substitutes has increased as a result of this environmental concern.

Despite these benefits, reliably predicting the compressive strength of RHA concrete is difficult, as the reinforced mechanisms between factors such as mix ratios, curing regimes, and inherent material inhomogeneity form a complex linkage. Considering the diverse characteristics of RHA concrete, statistically intensive analyses are required to effectively capture complex relationships and trends. Conventional predictive methods occasionally have difficulty coping with these complexities, which call for sophisticated strategies. In this scenario, machine learning (ML) has arisen as a great alternative, dealing with complex data patterns and achieving high prediction power while minimizing the time- and cost-consuming experimental work [6]. Machine learning algorithms can learn from

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data independently, discovering the relationships among variables without a particular human intervention [7, 8]. Advanced machine learning methods have been effectively used in structural health monitoring of bridges [9], dynamic analysis of water tanks [10], asphalt behavior [11], self-compacting concrete [12, 13], corrosion in concrete [14, 15], and blended concrete mechanical properties [16–28].

Amongst machine learning methods, Artificial Neural Networks (ANN) are computational models inspired by the human brain's neural structure, and work by using interconnected networks of nodes that pass on the information. This model helps capture nonlinear relations and involves many variables without having previously defined their relationship [29, 30]. However, high-performance ANN needs an optimal parameter tuning, which is engaged and may not be actual. Genetic Algorithms (GA) solve that by iteratively mutating, selecting, and crossing over candidate solutions based on the principles of natural selection. The genetic algorithm takes a population-based search approach, making it ideal for exploring the multidimensional space of parameters to find a global optimum economically [31]. The construction of the ANN-GA hybrid neuro-genetic algorithm integrates the predictive power of ANN and the optimization capacity of GA. It is used to model the compressive strengths of RHA concrete. The concrete properties look simple at the macro level, but the meso-level strength development is caused by the multi-scale interaction of constituent materials. At the same time, methods that are statistically intensive in nature such as ANN-GA that are data-driven, are in a good position

Many studies have used machine learning for predicting concrete compressive strength [32]. However, models show a fair degree of variability due to the diversity of input parameters and optimization algorithms. GA can be configured and optimized for core algorithm architectures. This enhances ANN's forecasting performance. The predictive performance of the ANN-GA hybrid model also holds promise for RHA concrete multivariate prediction models [33]. Current optimization methods like gene expression programming (GEP), particle swarm optimization with differential mutations (PSOTD), particle swarm optimization (PSO), and biogeography-based optimization (BBO) have shown promise. Still, their effectiveness may differ depending on circumstances. This investigation fills the gap existing in research by proposing a comprehensive GPR model based on ANN-GA, which is dedicated to RHA concrete, and evaluates the reliability of predictions for salient input and output relationships. Using RHA to develop such a model can enhance scientific knowledge and eco-friendly engineering practices.

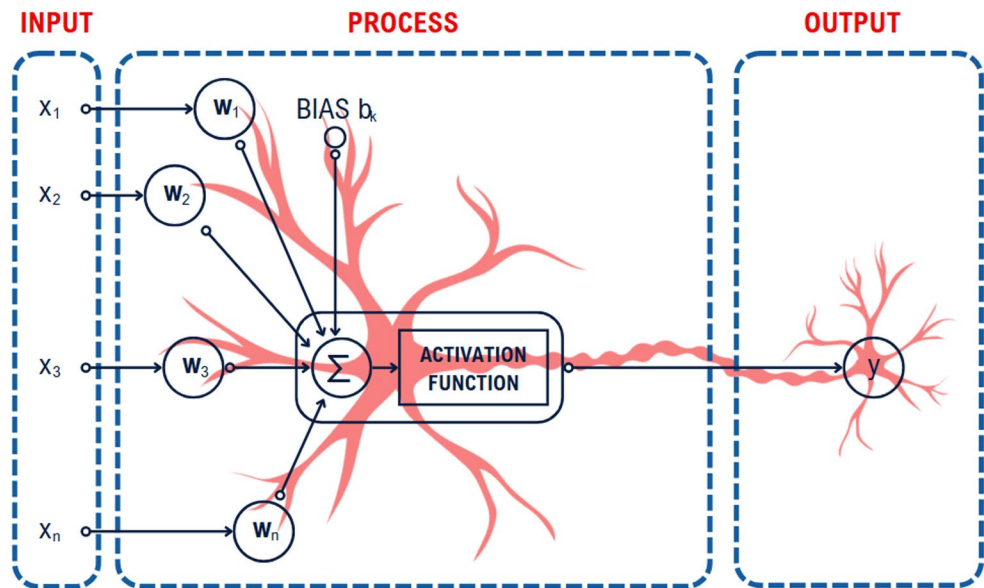
2 Research Significance

Although many researchers have examined the effectiveness of Rice Husk Ash (RHA) as a supplementary cementitious material in concrete, there is a significant gap in the literature around a multifaceted parametric study exploring the key variables that affect the compressive strength of RHA concrete. Most studies focus on single variables or preliminary experimental studies, and do not investigate in detail the interactions between the mix components (water, cement, aggregates, and RHA). The interactions of the RHA concrete components play a vital role in optimizing a mix and producing the required physical and mechanical properties of the concrete, particularly when using alternative materials like RHA. To address this gap in the literature, the current study introduces a hybrid predictive model based on an Artificial Neural Network (ANN) that is optimized using a Genetic Algorithm (GA). The predictive model accurately predicts the compressive strength of RHA concrete while simultaneously enabling an in-depth examination of inter-relationships amongst the input parameters. The current study features parametric and sensitivity analyses that help determine the most influential variables and their influence at different levels. These analyses contribute to a deeper understanding of the mix design process and demonstrate the effectiveness of machine learning techniques as a more efficient and cost-effective alternative to traditional experimental methods, especially in resource-limited settings.

3 Hybrid Neuro-Genetic Algorithm

Artificial Neural Networks (ANNs) are computational systems inspired by the human brain, consisting of interconnected neurons that process data and perform specific tasks [34]. Each neuron receives multiple inputs, applies weights and biases to influence connection strength, and produces a single output. The typical ANN architecture includes an input layer, one or more hidden layers, and an output layer, enabling it to model complex input-output relationships. Neurons use activation functions to determine whether signals should be passed to the next layer. Common ANN algorithms include the feedforward algorithm, where data flows one way from input to output without feedback loops, and the backpropagation algorithm, which adjusts weights by propagating errors backward to improve accuracy [35]. These features allow ANNs to detect patterns, learn from data, and provide accurate predictions, making them highly effective for solving nonlinear and complex problems across various domains. Figure 1 illustrates the flowchart of the ANN, outlining the key processes involved in the network's structure and operation. As stated by Jalal, et al. [36] in their

Fig. 1 Schematic imitation of a biological neuron by an artificial neural network



study, the complex behavior of rubberized cement composite was captured by using ANN, indicating that this type of machine learning method is effective when utilized in non-linear relationships. While ANN is an accurate prediction modeling machine, it must be trained under various conditions to ensure precision and accuracy in different cases [37]. To ensure that ANN produces reliable results, this is often fine-tuned with varying optimization techniques such as PSO, WOA, and GA [38].

ANN is non-linear and solving an optimization problem with exact methods is found to be hard. In this structure, the choice of input variables affects the accuracy directly and the number of neurons in hidden layers cannot be manipulated to ensure the optimal units of neurons. To address this, metaheuristic algorithms are utilized to provide solutions for ANN optimization [39]. GA is often used to optimize neural networks and solve problems without pre-defined efficient solutions by providing optimal alternatives for calculations [40]. The flowchart in Fig. 2 illustrates the integrated process of ANN and GA through the combined model training, optimization, and prediction method. The training of the ANN with GA outperformed that of other models. As stated by Liu et al. [41], the results of ANN, ANN-GA, and other hybrid models comparing the load-settlement behaviors of eco-friendly raft-pile foundation systems indicated that ANN-GA would outperform other models. In the research conducted by Santhosh et al. [42], the researchers recommended using ANN and GA models to create highly predictable results of alloy steel components. A challenge often met in developing machine learning models is the errors of underfitting or overfitting, which subsequently impact the model's performance. Overfitting occurs when a model identifies the actual patterns of the training data, other noise, and random variations, resulting

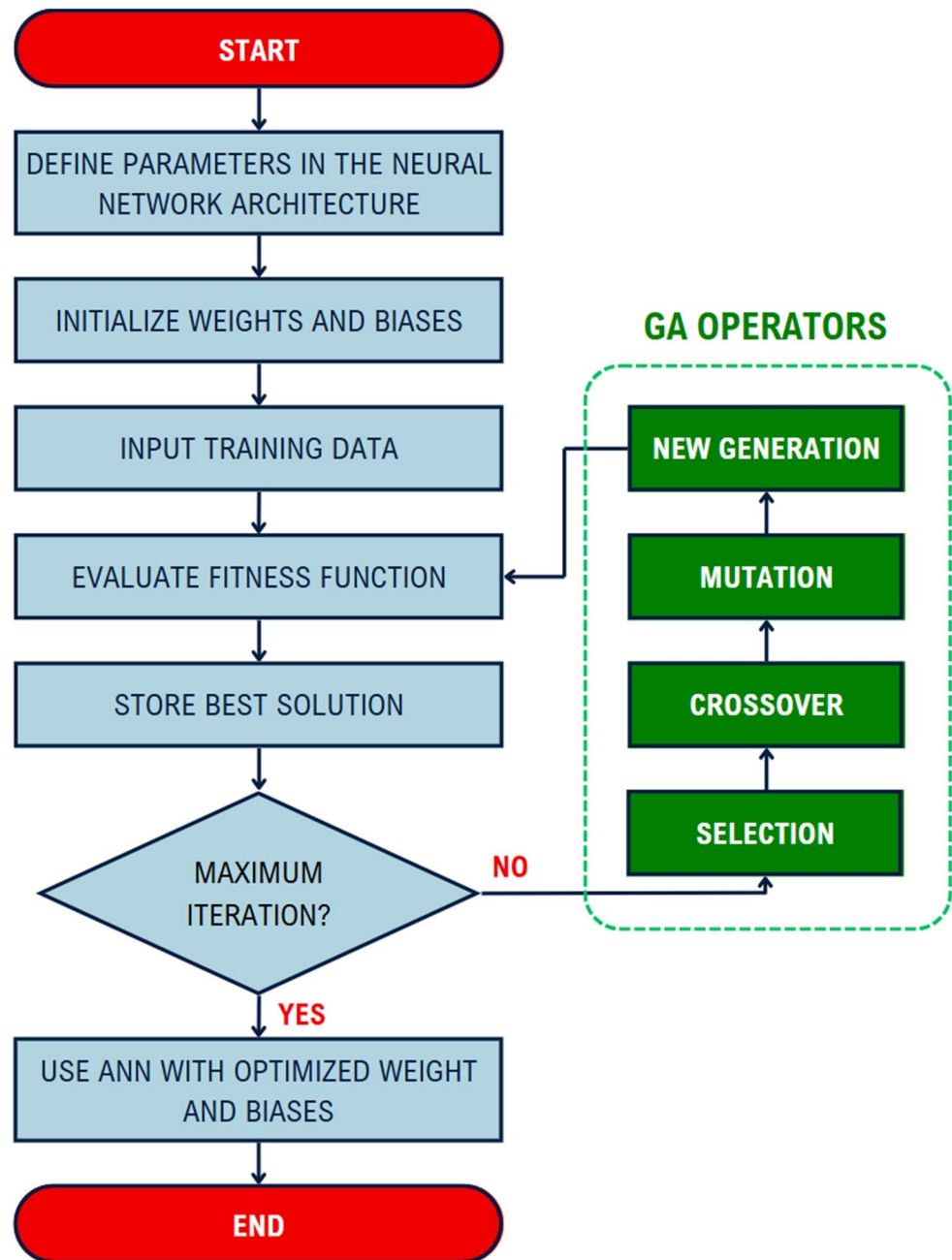
in a model that does not generalize to new unseen data and has less predictive reliability. Underfitting occurs when the model is too simple to learn the underlying form of the data, resulting in a less accurate model in training or testing datasets. To effectively mitigate these problems, several well-established techniques can be applied. Cross-validation is a technique used to assess model performance based on multiple 'folds' or subsets of the dataset, ensuring the model results are stable and consistent regardless of the subset used. Early stopping prevents overtraining by stopping the learning process when performance on the validation set no longer improves. Regularization penalizes large weights or complexity to reduce the chance of overfitting. Generally, in neural networks, dropout turns off random neurons during training, allowing the model not to rely solely on a specific path. Data augmentation artificially increases the size of the training set by creating modified copies, leading to knowledge generalization. When all of these strategies are used correctly, the capacity of the model to learn usable patterns is improved. At the same time, constraints on both overfitting and underfitting are reduced for more accurate, and robust predictions.

4 Data Analysis

4.1 Data Statistics

Table 1 shows an overview of the studies that contributed to the data collection phase, including reference citations and the number of data points from each publication [3–5, 43–80]. This is provided as open and transparent documentation of the data collection stage, and where the dataset was obtained for the development and analysis of models. Each

Fig. 2 Optimization procedure of the defined artificial neural network using genetic algorithm



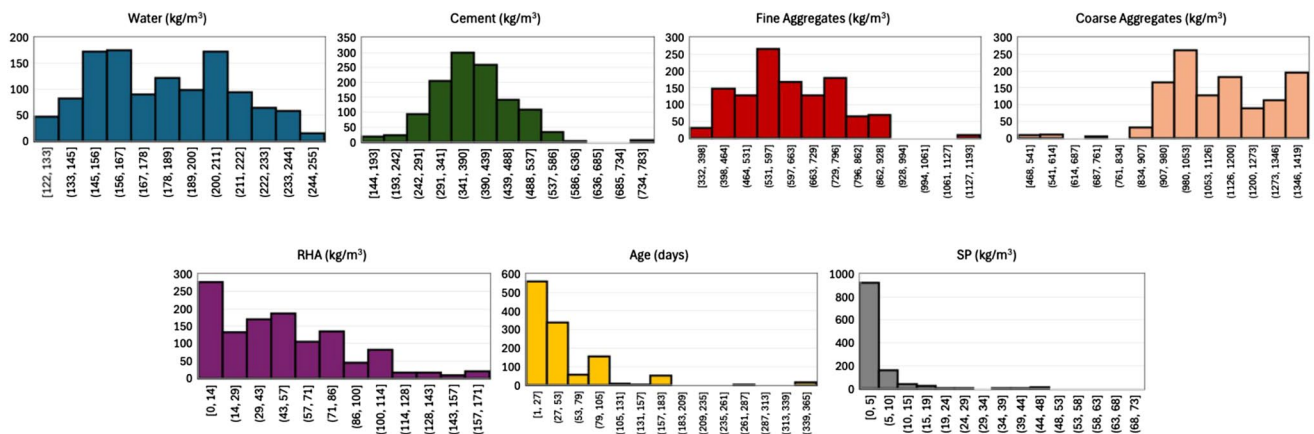
citation in the table identified the published study contributing to the dataset in accordance with the inclusion criteria in the systematic review phase. The number of individual data sample entries taken from each publication is also displayed, indicating the relative contribution of the study to the final dataset. The display of citations methodically outlines where the data were sourced and also provides replicability of the research for other researchers. A variety of data sources is also shown, which can provide important context for the validity of the developed predictive models. By showing the number of sample entries made per citation, it

also indicates possible bias from studies disproportionately contributing to the final dataset for each data sample.

Figure 3 illustrates the distribution of key parameters within the consolidated dataset used in this study. The figure reveals that the variables exhibit broad distributions, indicating that the dataset encompasses a wide range of experimental values. This diversity is crucial for developing a robust and generalizable predictive model. A dataset with wide parameter variation allows the model to learn from different combinations of input features, thereby improving its ability to predict the compressive strength of Rice Husk Ash (RHA) concrete under various conditions. Such variability

Table 1 Sources of data distribution

Reference	Number of data	%	Reference	Number of data	%
Sata et al. [3]	20	1.68	Khassaf et al. [43]	16	1.34
Mahmud et al. [4]	60	5.03	Abalaka [44]	140	11.74
Mahmud & Hamid [5]	16	1.34	Rodríguez De Sensale [45]	27	2.27
Singh [46]	10	0.84	Hwang & Chandra [47]	24	2.01
Singh & Res [48]	35	2.94	Zhang & Malhotra [49]	14	1.17
Chao-Lung et al. [50]	42	3.52	Islam et al. [51]	60	5.03
Ferraro & Nanni [52]	9	0.76	Buiet al. [53]	60	5.03
Ganesan et al. [54]	76	6.38	Sakr [55]	20	1.68
Noaman et al. [56]	5	0.42	Gautam et al. [57]	24	2.01
Olutoge & Adesina [58]	36	3.02	Nisar & Bhat [59]	15	1.26
Ramezaniapour et al. [60]	16	1.34	He et al. [61]	8	0.67
Krishna et al. [62]	13	1.09	AkshayTandon [63]	5	0.42
Siddika et al. [64]	27	2.27	Quing-Ge et al. [65]	28	2.35
Ikpong & Okpala [66]	48	4.03	Zareei et al. [67]	12	1.01
Ramasamy [68]	50	4.19	Habeeb & Bin Mahmud [69]	20	1.68
Kishore et al. [70]	24	2.01	Madandoust et al. [71]	20	1.68
Saraswathy & Song [72]	21	1.76	Habeeb & Fayyadh [73]	24	2.01
Kartini et al. [74]	18	1.51	Chopra et al. [75]	12	1.01
Muthadhi [76]	76	6.38	Ismail & Waliuddin [77]	16	1.34
Anwar et al. [60]	18	1.51	Zerbino et al. [78]	12	1.01
Givi et al. [79]	15	1.26	Total	1192	100

**Fig. 3** Parameter distributions**Table 2** Data statistics of the input parameters

Parameters	Water (kg/m ³)	Cement (kg/m ³)	FA (kg/m ³)	CA (kg/m ³)	RHA (kg/m ³)	Age (days)	SP (kg/m ³)	CS (MPa)
Mean	181.06	387.63	628.83	1127.29	48.40	39.93	3.48	49.56
Median	180	382.5	612	1113	46	28	0	44.82
Mode	165	425	446	1419	0	28	0	52
Sample Variance	918.41	7478.35	20533.76	33166.54	1560.78	3258.29	61.28	623.96
Standard Deviation	30.29	86.44	143.24	182.04	39.49	57.06	7.83	24.97
Minimum	122.4	144	332	468	0	1	0	2.35
Maximum	255	783	1193	1419	171	365	72.6	143.78

ensures that the model can be applied effectively across different concrete mix designs.

Table 2 presents a summary of the statistical data for each parameter, including measures of central tendency, sample

variance, standard deviation, as well as the minimum and maximum values. The arithmetic mean values of the seven parameters across the combined 1192 datasets are as follows: water at 181.062 kg/m³, cement at 387.629 kg/m³, FA

Table 3 Models in predicting compressive strength of RHA concrete

Authors	Formulas
Li et al. [26]	$f_C = -0.47317 + 0.297 (Age) + 0.0779 (Cement) - 0.0732 (RHA) - 0.145 (Water) + 1.524 (SP) + 0.0154 (Aggregate)$
Al-Hashem et al. [32]	$CS = A + B + C + D + E$ where: $A = Water - \frac{11.67}{2 \times F.ag. - SP - 2 \times C.ag. - 0.24 + RHA}$ $B = ((Cement \times (-1.02 + Age) + SP)^{\frac{1}{3}})$ $C = \frac{F.ag.}{(-8.37 - RHA) + (Age \times Water) - (C.ag. + SP + d4)} - Water$ $D = \frac{((Age \times SP) + (12.37 \times Cement) - (42.51 \times FA)) - Age}{Water}$ $E = (1.59 - (\frac{Age \times SP}{9.49 - RHA}) \times (-8.39 + RHA) + 9.49)^{\frac{1}{3}}$
Amin et al. [81]	$f_C = \frac{W}{D\sqrt{Z}} \sqrt{Z\sqrt{D} - D} + X - \frac{X}{D}$ where: $Z = C + RHA - W - \sqrt{D}$ $X = \frac{FA}{W + \sqrt{D} - SP\sqrt{Z}}$
Saridemir et al. [82]	$f_C = \left[\left(\sqrt{(AS + 7.24)} + \sqrt{(AS * SP) + AS} \right) + SP \right]$ $* \left[\frac{PC30 - \left(\frac{\left(\frac{W}{AS} * 1.54 \right)}{-9.92} \right)}{A} - 2.74 \right]$ $* \left[\left(\frac{(PC30 - (-9.60 * AS)) - (RHA - PC40) - PC30}{A} \right) - 2.74 \right]$

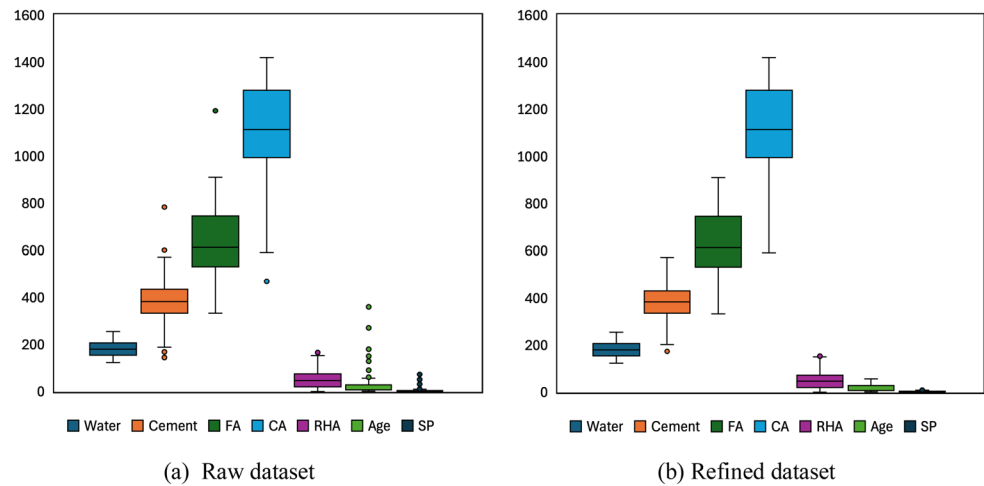
at 628.826 kg/m³, CA at 1127.290 kg/m³, RHA at 48.397 kg/m³, age at 39.932 days, SP at 3.479 kg/m³, and the actual compressive strength at 49.558 MPa. The median values for these parameters are 180 for water, 382.5 for cement, 612 for FA, 1113 for CA, 46 for RHA, 28 for age, 0 for SP, and 44.815 MPa for compressive strength. The closeness of the mean and median values suggests that the dataset is relatively symmetrically distributed. As shown in Table 3 are the empirical equations used to predict the compressive strength of RHA.

Outliers are data points that significantly differ from the general trend of a dataset, either being much higher or lower than other observations. These outliers affect the overall statistics of the data, such as the mean and standard deviation, providing a biased analysis and possibly a flawed conclusion. Box plots are used to identify and visualize outliers, as shown in Fig. 4. A box plot (or box-and-whisker plot) is a reasonably simple display to interpret by utilizing five summary statistics to visualize data distributions and variability, namely: minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum. The interquartile range (IQR) is a statistic used to identify outliers using the standard way of identifying outliers at 1.5 × IQR. Any point that fell below

Q1 - [1.5 × IQR] or above Q1 + [1.5 × IQR] is considered an outlier and appeared as a point isolated from the rest of the data beyond the plotted whiskers, as shown in Fig. 3a. To minimize impacts from detected outliers and retain the statistical properties of the dataset, detected outliers are replaced with the median value associated with each of the individual parameters. This imputation option allowed us to preserve the central tendency of the dataset while curtailing the impact of extreme values. The decision to use median substitution is made to retain potentially useful information, as extra values are especially well-represented in some variables [24, 25]. While this method maintains data continuity, other techniques, such as robust scaling or winsorization, may provide even more robustness for data with skewed distributions. The new box plot after outlier treatments is shown in Fig. 3b.

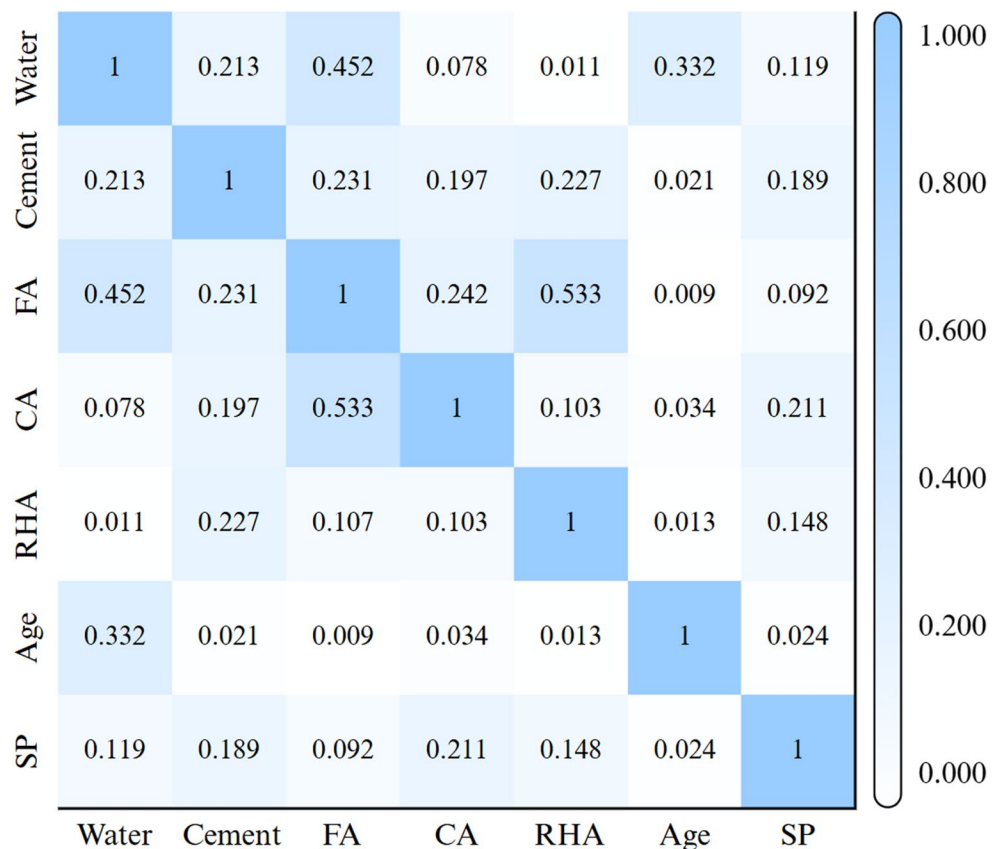
4.2 Mutual Information Heatmap

To provide further analysis and insight into the dependencies among the seven input parameters in the model, a Mutual Information (MI) heatmap is developed as illustrated in Fig. 5. This method provides a broader perspective

Fig. 4 Elimination of outliers in the dataset

on the strength and nature of relationships between the variables, covering linear correlations and nonlinear interactions too. The MI values suggest a moderate dependency between water and cement. This relationship highlights the importance of the water-to-cement (w/c) ratio in developing concrete strength. A low but unique mutual information score is observed between water and RHA, indicating a weak dependency whereby an increase in RHA will demand additional water content, especially if the superplasticizer (SP) dosage is limited. This is due to the surface area of the RHA and porous matrix that requires more water to

maintain workability and enable strength gain. A moderate dependency is also observed between RHA and cement values, showing the inverse relationship. This supports RHA as a suitable sustainable alternative to lessen the volume of cement used while helping with societal environmental goals without compromising the integrity or durability of the structure. Other observed relationships are a resemblance between FA and CA and an extremely weak or insignificant association between SP dosage and concrete age, suggesting very little direct association between the two variables.

Fig. 5 Mutual information heatmap

5 Architecture and Performance of the ANN-GA Model

5.1 Performance Outputs of the Different ANN-GA Model Architectures

The hybrid ANN-GA model created for estimating the compressive strength of RHA concrete utilized a three-layer feedforward neural network represented as $7-n-1$, which entailed an input layer, a hidden layer, and an output layer. The ANN-GA model is developed and trained using MATLAB R2015a with a Neural Network Toolbox and GA Optimization Toolbox on a macOS Monterey 12.6 system. The neural network - genetic algorithm is set up using the designed control parameters in Table 4. The 7-parameter input layer is water (kg/m^3), cement (kg/m^3), FA (kg/m^3), CA (kg/m^3), rice husk ash (RHA) (kg/m^3), age (days), and superplasticizer (kg/m^3). The hidden layer acted as an intermediary, where each neuron performed a weighted sum of the input values and added a bias and activation function before the output was passed along to the next layer. As data is collected from multiple datasets, ensuring consistency and comparability across all datasets is critical in the modeling process. A complete dataset alignment strategy was implemented to address this step. First, it required verifying that every variable performed across the various datasets is consistent in definition, representation, and relevance to the model. Moreover, data normalization is systematically used so that all numerical values can be adjusted to a standard scale, removing discrepancies caused by having problems with different units or magnitudes. This process allowed for the accurate comparison and integration of data, enabling the machine learning model to learn from a unified and standardized input structure.

To ensure that the model was reliable and generalizable across different dataset segments, the model's development process involved using a k-fold cross-validation method.

Table 4 Control parameters of the neuro-swarm algorithm

Parameters	Compressive Strength
Number of Input Layer Nodes	7
Number of Hidden Nodes	1–50
Learning algorithm	Feed forward
Training function	Levenberg-Marquardt
Activation function	Hyperbolic tangent sigmoid
Learning Rate	0.001
Population size	100
Epochs	500
Parent Selection	Roulette wheel
Generic operator	Cross over, Mutation
Cross over rate	0.90
Mutation rate	0.05
Performance indicator	MSE, R

This approach minimizes overfitting and produces a more accurate estimate of the model's predictive ability by systematically rotating the data used for training and testing. In this instance, the data was treated in a manner that allowed for state-proportional representation of the data sources across both training and testing sets, meaning that each source was organized around a consistent 70:30 ratio. Because each source was split in this manner, this helped to mitigate potential biases that may have been present through splitting the data in diverse ratios; thereby ensuring that each training set was balanced equally to support as many diverse learning environments and evaluations as possible. In this approach, data was split into multiple folds, allowing the model to be trained and validated 10 times, using a different fold as a validation set, and a collection of the rest as a training set. This ensures that each data point is used in both training and validation phases, thereby improving the robustness and fairness of the evaluation. To mitigate the risk of overfitting and design a model with more generalized capacity, a number of techniques were employed in the development of the model. These included early stopping, which monitors the model's performance on a validation set and halts training once performance no longer improves, thus preventing overtraining. Regularization was also applied by introducing a penalty term to the function during training, effectively limiting the model's complexity. This approach discourages the model from becoming overly dependent on individual features or weights, thereby helping to reduce the likelihood of overfitting and improve generalization to unseen data. A validation phase was also employed to evaluate the model's performance on new unseen data throughout training. Furthermore, a large dataset was utilized, and the model complexity was reduced by limiting the number of layers used in the architecture. This combination of strategies was implemented to ensure that the final model would not only fit the training data well but also perform effectively on new, unseen datasets, thereby enhancing its robustness in practical applications. To identify the optimal architecture, the model was trained using varying numbers of hidden neurons, ranging from 1 to 50. Each configuration was trained for up to 50,000 iterations, and its performance was evaluated based on two key performance indicators: the mean squared error (MSE) and the correlation coefficient (R). For each architecture, its corresponding final weights and biases were also recorded.

The Genetic Algorithm (GA) was utilized to fine-tune and optimize the initial weights and biases of the neural network. Its process is presented in the lower right section of Fig. 6. GA mimics the process of natural selection by evolving a population of candidate solutions, where each represents a unique set of weights and biases. The initial population is generated randomly, and each is evaluated

Fig. 6 ANN-GA 7–36–1 model architecture of the RHA concrete compressive strength model

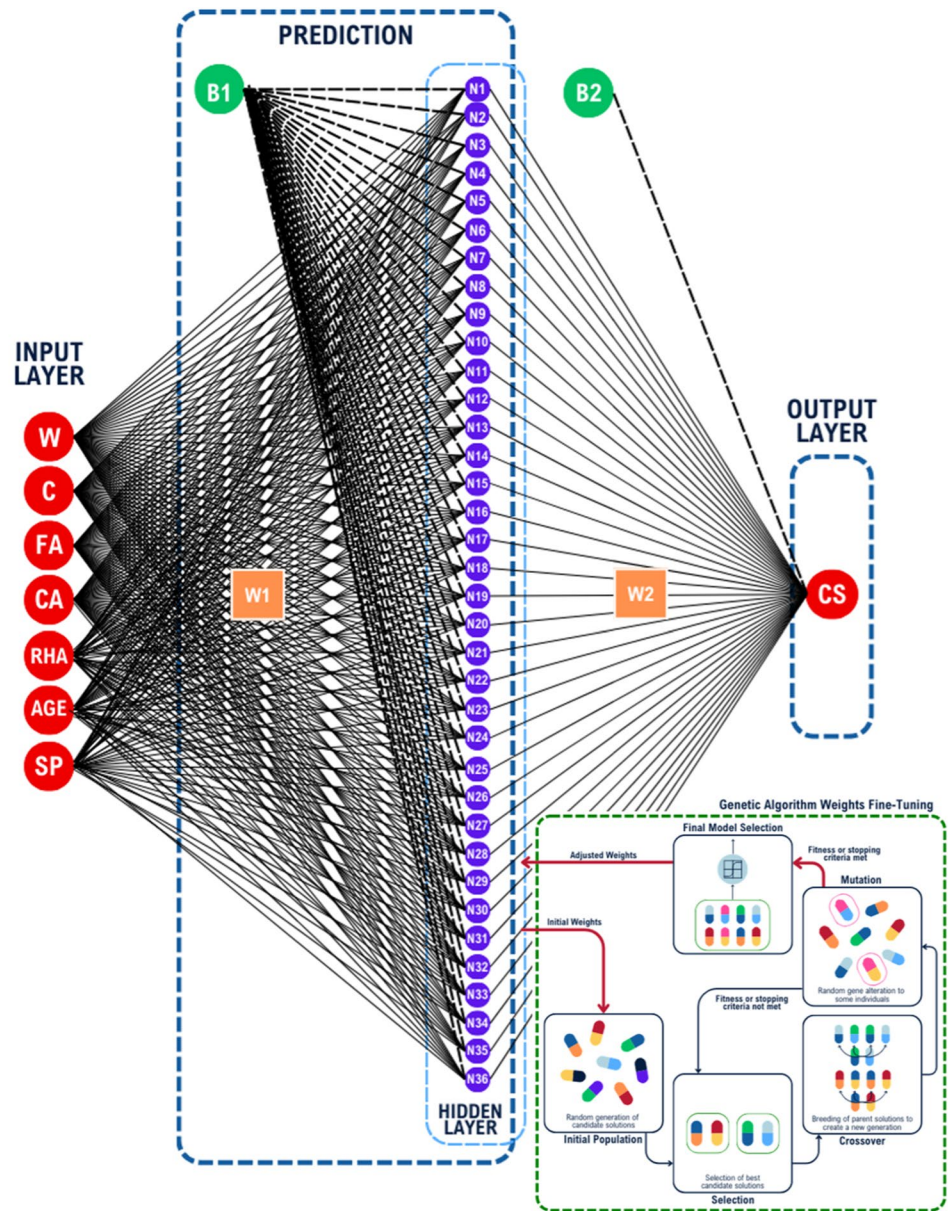
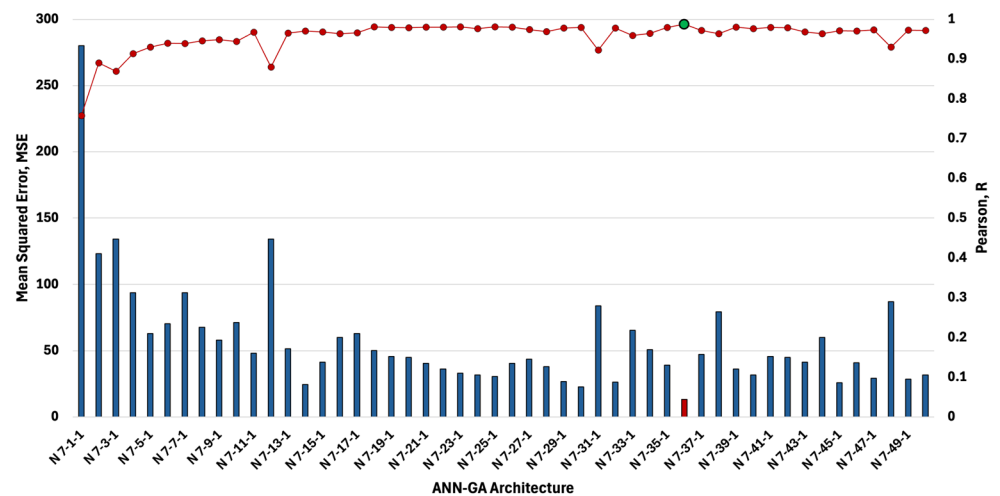


Fig. 7 Graphical representation of the correlation coefficient (r) and mean squared error (MSE) values of the different ANN-GA architectures



based on a fitness function utilized by the ANN, MSE. The best-performing individuals are selected and undergo genetic operations such as crossover and mutation to produce a new generation of solutions. This iterative process enables the algorithm to effectively explore the solution space and avoid premature convergence. Ultimately, the optimized set of weights and biases enhances the ANNs learning capability by minimizing the risk of getting trapped in local minima, thereby improving the model's predictive accuracy for compressive strength.

The final architecture was chosen for having the lowest MSE and highest R. Figure 7 shows performance trends across the different architectures. As a result, the 7–36–1 architecture was adopted as the general model for this study, achieving the lowest MSE (13.225) and the highest R (0.988). It has an input layer of 7 neurons for the seven input parameters, a hidden layer of 36 neurons, and a single output neuron representing the predicted compressive strength of RHA concrete. Using k-fold cross-validation, the model was evaluated across multiple folds to check robustness and generalizability. Mean values of the key metrics across all folds were: $R=0.979$, $MAE=7.885$, $RMSE=8.922$, and $MAPE=10.337$. With a 95% confidence interval, variation across folds stayed within acceptable limits: ± 0.011 for R, ± 5.332 for MAE, ± 6.224 for RMSE, and ± 5.801 for MAPE. These results demonstrate a high level of consistency in the model's predictive performance. When applied to the locked test set, the model yielded corresponding values of 0.975 for RR, 7.886 for MAE, 8.322 for RMSE, and 12.215 for MAPE, confirming the model's strong generalization capability.

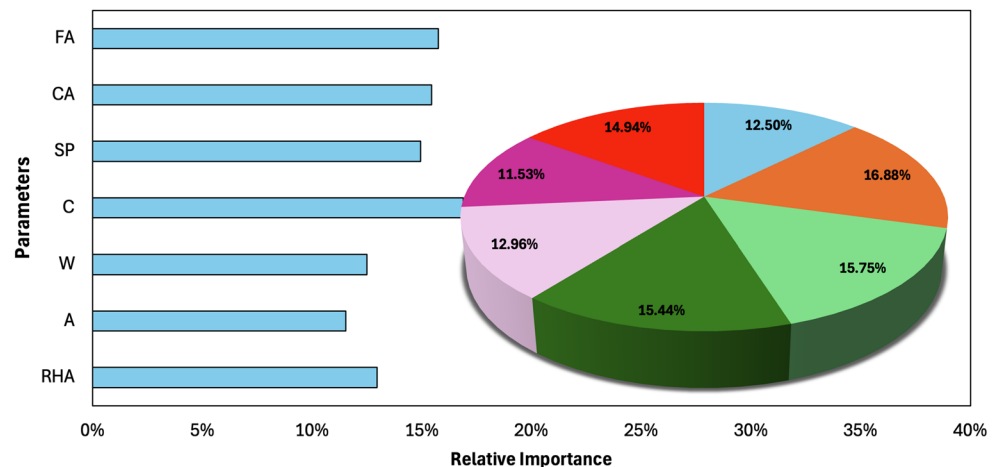
5.2 Weights and Permutation Importance of ANN-GA 7–36–1 Model

A permutation importance analysis is utilized to assess the relative contribution of each input variable in estimating the compressive strength of rice husk ash concrete using

the proposed ANN-GA model with a 7–36–1 configuration. The underlying principle is to shuffle the values of each input feature consecutively while preserving the values of the other features. The reshuffled dataset is then fed into the trained neural network model to generate predictions. Since the shuffling breaks the true relationship between the input and output, a drop in prediction accuracy is expected if the shuffled feature is essential. The primary metric used for performance is the root mean squared error (RMSE). The difference in RMSE between the original test set and the shuffled test set is calculated and expressed as a percent difference to represent the importance of that feature. The above steps are repeated independently for all features to maintain fair, unbiased comparisons of importance. The results indicate that cement is the most significant predictor of compressive strength, resulting in a 16.88% increase in RMSE when shuffled. Fine aggregate and coarse aggregate are closely followed, with respective importance values of 15.75% and 15.44%. Interestingly, RHA, despite being the central material of interest in this study, showed a lower relative importance of 12.96%.

To better visualize the proportional contribution of each parameter, Fig. 8 illustrates the results of the analysis. Cement plays a central role in the development of concrete strength as it serves as the primary binding material in the mix. When combined with water, it undergoes a hydration process that leads to the formation of calcium silicate hydrate (C-S-H), the key compound responsible for the hardening and strength gain of concrete. The amount and quality of cement significantly influence both the early and long-term compressive strength. Aggregates, comprising approximately 60–80% of the concrete volume, form the structural framework or “skeleton” of the mix. Their properties—including size, shape, texture, grading, and strength—affect the density, compaction, and load distribution within the concrete, thereby influencing its mechanical performance. Meanwhile, superplasticizers are essential chemical admixtures that enhance the workability of concrete without

Fig. 8 Permutation importance of the input parameters in RHA concrete



the need for additional water. By dispersing cement particles more effectively, they reduce the water-cement ratio, which is a critical factor in achieving high compressive strength. Superplasticizers also promote better compaction and reduce porosity, resulting in a denser and more durable matrix. These three components—cement, aggregates, and superplasticizer—emerged as the most influential variables in predicting concrete compressive strength because of their direct impact on the mix's hydration kinetics, microstructure formation, and overall mechanical integrity. Their dominance in various predictive models highlights their fundamental role in strength development, especially in conventional and high-performance concrete mixtures. The high importance of aggregates, particularly fine aggregates, was attributed to their role in concrete packing density and microstructure: achieving better homogeneous, workable, and durable mechanisms, which directly influence the compressive strength of concrete. Kiambigi et al. [83] underlined the impact of fine aggregates' physical, chemical, and mineralogical properties on improving the compressive strength of concrete. Moreover, fine aggregates impact multiple non-linear interactions within hidden layers. The multidimensional effect of fine aggregates apprehended high synaptic weights due to its particle interlocking mechanisms and flowability properties.

Coarse aggregates and their load-bearing capacity characteristics (size, angularity, and interfacial transition zone) allow them to be the second most influential parameter in enhancing compressive strength as they withstand propagation and boost stiffness. Ogundipe et al. [84] examined the significant influence of coarse aggregates size. As a result of their experiment, as the coarse aggregates increases to 12.5 mm, so does the compressive strength, and its peak performance is at 20 mm. However, a substantial drop in strength was observed exceeding this size, particularly at 25 mm. Following that, although the influence of coarse aggregates is slightly less than fine aggregates in this model, the ANN-GA model acknowledges its stabilizing effect on the output parameter through constant weight contribution across neurons. With the fourth highest influence rating, the SP is responsible for reducing water demand through the dispersion of cement particles by electrostatic and steric mechanisms. Khan and Ali [85] underscored the influence of SP to enhance the compressive strength of concrete. Their study establishes that by incorporating SP up to an optimal dosage of 1% by weight of cement, the 28-day compressive strength exhibits the best performance. Therefore, SPs non-linear relationship with strength improves the strength up to a particular optimum dose, and past that, it yields segregation. The model apprehends influential weighted paths, establishing SP as a highly significant parameter in the hidden layer dynamics. The principal hydraulic binder,

cement, provides structural integrity. For concretes with RHA-blended systems, the cement provides the calcium necessary for the pozzolanic reaction, resulting in early and secondary strength development. Eskandari-Naddaf and Kazemi [86] conducted research verifying how paramount cement is as a parameter that influences compressive strength. By removing the cement strength class from the input features, the coefficient of determination (R^2) drops from 0.94 to 0.74. It is also imperative to consider, though critical, that the cement's direct influence decreases due to its partial replacement with RHA: this elucidates why the model diagnoses cement with a lower relativity than SP.

Water, on the other hand, streamlines the hydration and workability of cement mix. Generally, a lower w/c ratio leads to higher density and strength. However, inadequate water impedes the hydration process, and excess water forms internal voids, lowering the strength of the concrete. Since water shows no linear interaction with multiple inputs and is conditional on other parameters, it is safe to consider that its interdependency is why the hybrid ANN-GA model allocates a moderate PI% to it. Ikpa et al. [87] demonstrate the necessity of evaluating the quantity and quality of water in concrete mix design as they adversely influence cement hydration, which may enhance or compromise concrete strength. For instance, water free from impurities (distilled water) reached the highest average compressive strength of 17.96 N/mm², while secondary wastewater attained a lower strength of 12.09 N/mm². Their study endorses the results of the RI analysis, where water, although not as critical as the other parameters, occurred as a crucial variable controlling the strength of the concrete. The age of concrete directly affects its strength. In the case of RHA concrete, the strength increases at 28 days and beyond as amorphous silica reacts with calcium hydroxide. However, since 900 out of 1192 (75.38%) data samples are concentrated at early stages (7 to 28 days), the RI% shows that age is less impactful than composition in this dataset. Pourbaba et al. [88] revealed that the rate of hydration and microstructural densification, which are imperative factors in strength development, are time dependent. Based on the study reports, about 90% of their ultimate compressive strength is achieved by day 18 and nearly 100% by day 42.

RHA's value to the concrete strength leans on its highly reactive pozzolan characteristics that refine the pore structure and enhance long-term durability and strength. However, RHA stays time-dependent and susceptible to particle size, replacement level, and degree of fineness, resulting in the lowest RI% among the seven input parameters. Despite RHA's strength contribution being less direct in the context of the provided dataset, its significance still ought to be considered carefully. The study by Singh, R. and Singh, D. [46] accentuates that an improved compressive strength

of high-strength concrete is feasible by optimizing the replacement level of cement with RHA, as it maintains a suitable water-to-binder ratio. The RI% values revealed how different parameters interact with each other to influence compressive strength. While traditional components such as aggregates and admixtures demonstrate relevance in short-term strength behavior, supplementary cementitious materials (RHA) provide harmonious advantages over time. Through the hybrid ANN-GA model, the layered relationships were captured, and the weight of hybrid modeling techniques in advanced concrete design was highlighted.

6 Prediction Performance of the ANN-GA Model

6.1 Error Histogram of the ANN-GA Model

Figure 9 presents the error histogram of the training ($N=834$), validation ($N=179$), and test datasets ($N=179$) generated by the ANN-GA 7–36–1 model. The distribution of errors is evidently narrow and centered around zero, indicating that the model is well-trained with minimal deviation in predicting the compressive strength of RHA concrete. Approximately 94% of all predictions across the three data subsets fall within the $\pm 20\%$ error range, while the remaining 6% exceed this threshold. Notably, only 4 out of the 1192 data points exhibited errors greater than -60% , accounting for just 0.48% of the entire dataset. Given the extremely small proportion of these high-error instances, their overall impact on the model's performance is negligible, further reinforcing the robustness of the model's predictions. A confidence interval analysis was employed in this study to statistically estimate the range within which the

true compressive strength of concrete is likely to fall, based on the sample data. By applying a 95% confidence level, it implies that if the same sampling and analysis were repeated multiple times, approximately 95% of the calculated intervals would contain the true mean compressive strength. This approach provides a more informative interpretation than a single point estimate, as it accounts for the inherent variability and uncertainty in the data. In this case, the margin of error was determined to be ± 1.4 MPa, meaning that the true compressive strength is expected to lie within 1.4 MPa above or below the predicted mean value. This statistical range helps assess the reliability of the prediction model, enhances decision-making in concrete mix design, and supports the evaluation of whether the results meet acceptable engineering tolerances or performance specifications.

The relationship between the actual and predicted compressive strength of RHA concrete was visualized through a scatter plot, as shown in Fig. 10. All data subsets—training, validation, and test—exhibited strong predictive performance, with correlation coefficients (R) exceeding 0.98. The highest R -value was observed during the training phase ($R=0.988$), followed by the test phase ($R=0.989$), and then the validation phase ($R=0.986$). Meanwhile, the mean absolute percentage error (MAPE) measured at 5.632% or below across all phases, indicating high predictive accuracy. Interestingly, the ranking based on MAPE differed from that of R : the validation phase achieved the lowest MAPE of 4.221%, followed by the training phase at 6.043%, and the test phase at 6.153%. This variation highlights the difference in what each metric captures: while R measures how well the model captures the overall trend or correlation between actual and predicted values, MAPE reflects the average magnitude of relative error. Thus, the model has shown a strong linear

Fig. 9 Error histogram of the prediction performance of the ANN-GA Model

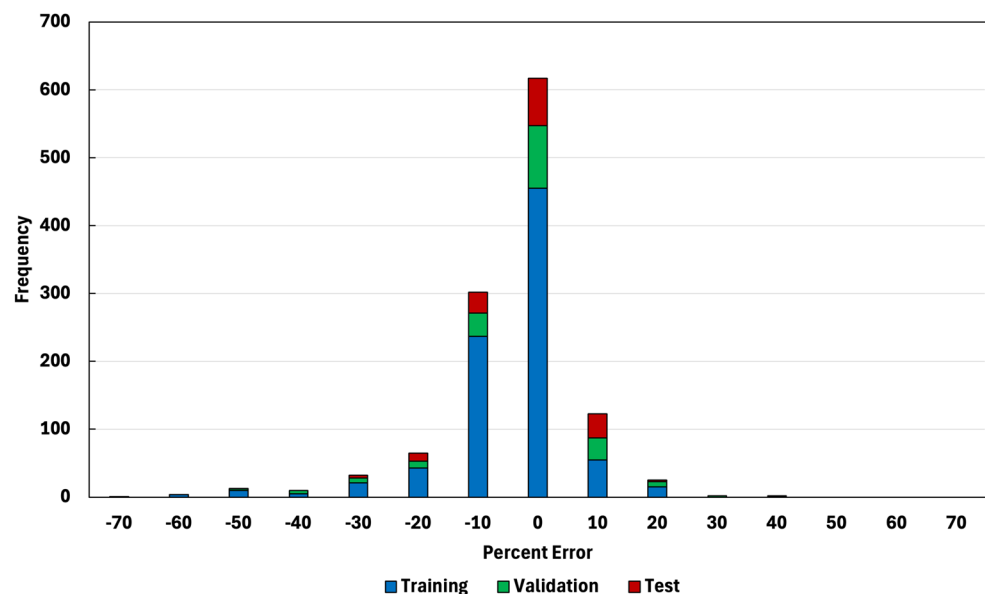


Fig. 10 Scatter plot of the prediction performance of the ANN-GA Model

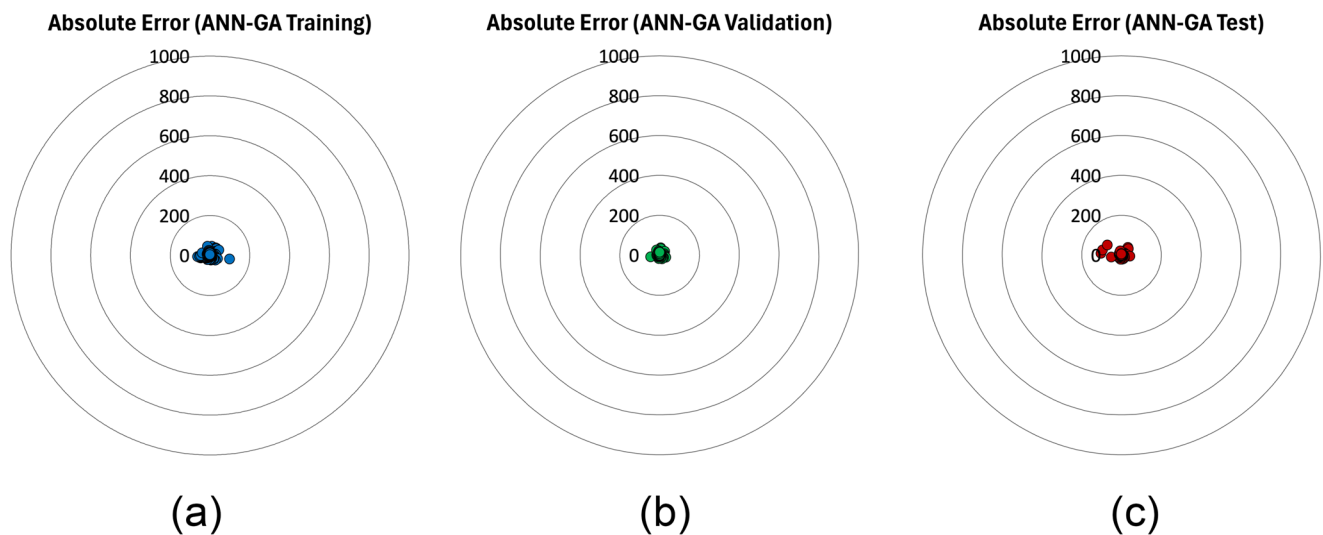
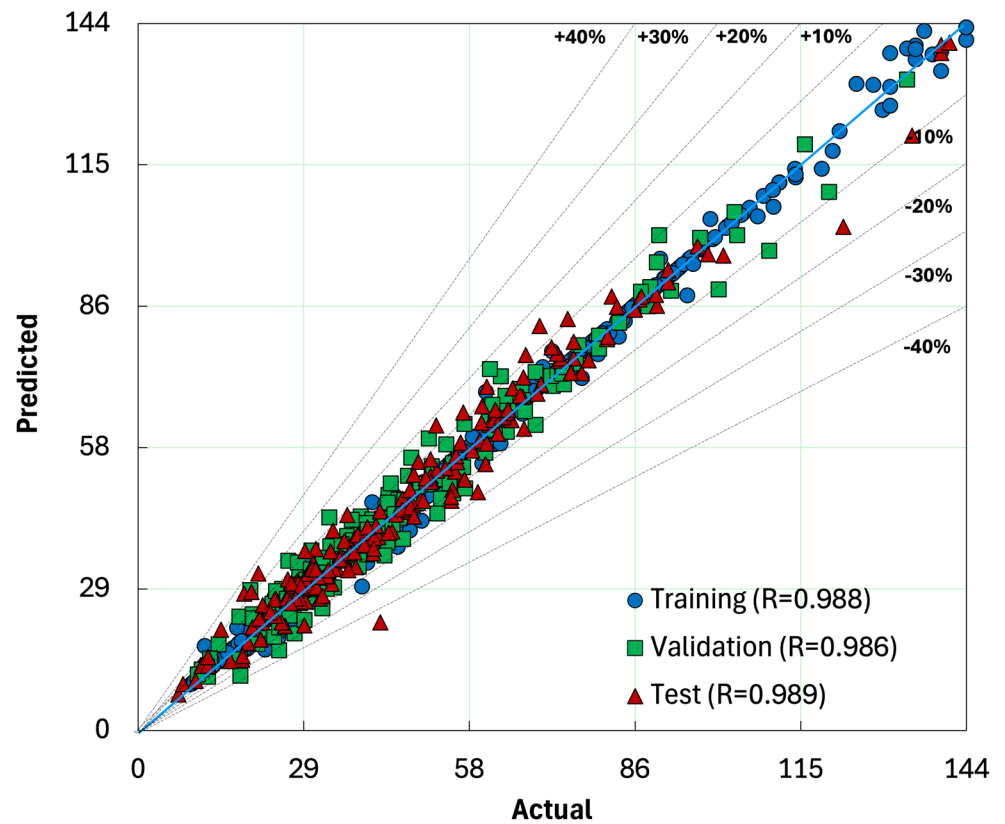


Fig. 11 Absolute errors of the ANN-GA model: (a) training, (b) validation, and (c) test subsets

relationship yet still has slightly higher prediction deviations, as observed in the training phase.

6.2 Absolute Error Analysis of the ANN-GA Model

Figure 11a presents a bullseye diagram illustrating the absolute error of the ANN-GA model across 834 training data points. The diagram features five concentric radial

lines, each representing a different absolute error threshold at 200% intervals. The innermost circle corresponds to errors within 0–200%. Notably, there was only one instance where the absolute error surpassed 100%, with a maximum of 108%, while another instance fell between 0 and 100%. The smallest absolute error recorded was 0.003%. Overall, the model achieved a mean absolute percentage error (MAPE) of 6.043% during the training phase. Figure 11b

demonstrated the absolute error of the ANN-GA model for the 179-validation data through a bullseye diagram. Within the validation dataset, there was no instance where the absolute error surpassed 200%. The smallest and largest value of absolute error recorded was 0.031% and 39.333%, respectively. Overall, the MAPE for the validation is 4.221%. While in Fig. 11c, a bullseye diagram was also utilized to show the absolute error of the ANN-GA across 179 test data points. The diagram also contained five concentric radial lines, each representing a different absolute error threshold also at a constant interval of 200. Within the testing dataset, there were three instances where the absolute error surpassed 100% namely 102.332%, 110.555 and 112.234%. The smallest absolute error recorded was 0.001%. Overall, the MAPE recorded was 6.153% for the test phase. Most of the errors are located within the first radial line. For the entire model, a MAPE of 5.632% was recorded, indicative of a great performance when it comes to prediction accuracy. The analysis indicates that the ANN-GA model performs effectively in predicting the values of compressive strength of RHA concrete with the validation phase having the lowest MAPE, followed by training phase and test phase, respectively. This suggests that the model can predict the values of new unfamiliar data during the test from the learnings in the training. Furthermore, there is a slight decrease in the value of MAPE from training phase to validation phase, which bounces back to a higher MAPE during the test phase. This suggests that the test phase of the data using ANN-GA model, although still precise, has shown a relatively higher prediction deviation as compared to the training phase. The performance of ANN-GA mode demonstrates a relatively high level of precision and effectiveness in predicting most of the target values in both training and test phase. This implies that the model can predict the

values of the compressive strength of RHA concrete during the simulation or test.

6.3 Stability Analysis, Residual Plot, and QQ Plot of the ANN-GA Model

Figure 12 illustrates the ratio of predicted to actual compressive strength values across the range of influencing parameters, normalized between 0 and 1. This ratio serves as an indicator of the ANN-GA model's stability in predicting the compressive strength of RHA concrete. The black horizontal line at a ratio of 1 represents perfect agreement between predicted and actual values. Data points clustered near this line signify stable and consistent model performance, whereas greater deviations reflect potential prediction variability. As shown in the figure, the majority of data points are concentrated around the ratio of 1. Notably, only 58 out of 8,344 points (0.695%) exceed a ratio of 1.5, with the highest recorded at 2.342. These outliers may be attributed to variations in testing apparatus, procedures, or experimental conditions across different studies. Despite this, the fact that most ratios fall within the range of 1 ± 0.5 reinforces that the ANN-GA model maintains stable and reliable prediction accuracy across a diverse dataset.

To evaluate the residual behavior of the proposed model and assess whether the underlying statistical assumptions hold, both predicted vs. residual plots and Q-Q (quantile-quantile) plots were generated using the test dataset. These diagnostic plots provide visual tools to examine the presence of bias, non-linearity, and heteroscedasticity in the residuals, as well as assess whether the residuals follow a normal distribution. Figure 13a presents the residual plot, where the residuals are plotted against the predicted values. The plot reveals a random scatter of residuals around the zero line, indicating that the model does not exhibit systematic bias

Fig. 12 Stability of the input parameters influencing the ANN-GA model

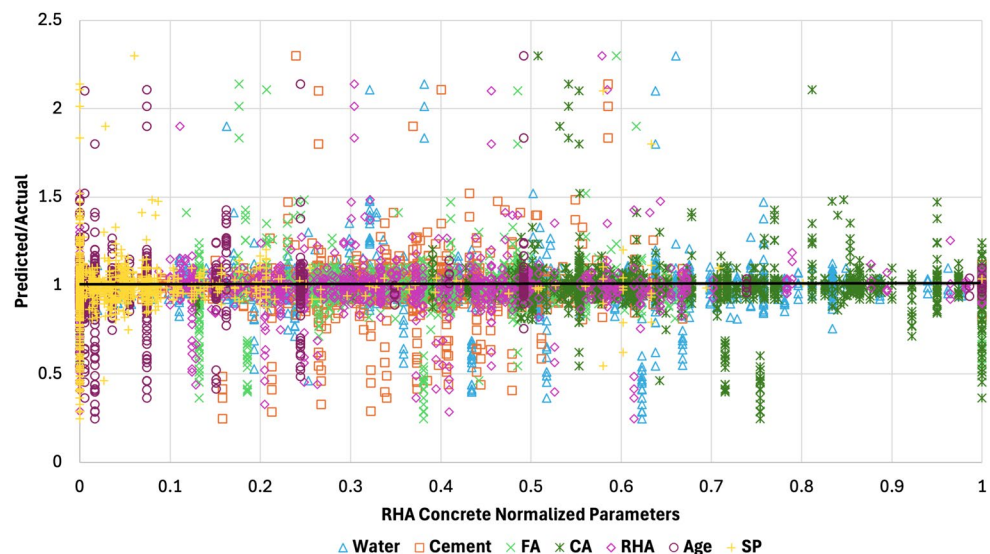
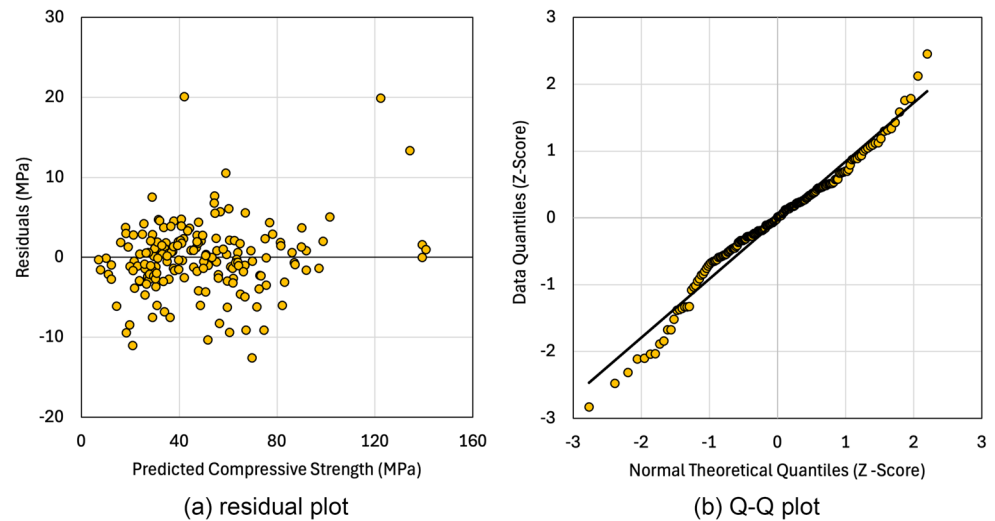


Fig. 13 Heteroscedasticity check

across the range of predictions. Furthermore, the absence of any visible fanning or funnel-shaped pattern suggests that the variance of the residuals remains constant, thereby ruling out heteroscedasticity. To further assess the distribution of residuals, a Q–Q plot was constructed, as shown in Fig. 13b. This plot compares the quantiles of the residuals to the quantiles of a normal distribution. The majority of the data points closely follow the 45-degree reference line, confirming that the residuals are approximately normally distributed. These findings support the validity and statistical reliability of the model's predictive behavior.

6.4 Model Comparison

For a direct, valid, and meaningful model comparison, each model must use the same and aligned data points for both predicted and actual values. However, due to the occurrence of negative, undefined, or imaginary outputs in the models, data filtering was necessary to exclude invalid results. To ensure transparency and reliability, the statistical measures for each model were calculated using their respective sets of valid data points, along with their corresponding Valid Prediction Ratio (VPR). This approach preserves the integrity of the analysis while accounting for differences in output quality across models. Out of the 1192 datasets, Saridemir et al. [82] model produced 1191 valid predictions, yielding a VPR of 0.999. The Al-Hashem et al. [32] model followed closely with 1190 valid predictions (VPR=0.998); the Li et al. [26] model returned 1188 valid predictions (VPR=0.997); while the Amin et al. [81] model returned 1177 valid predictions (VPR=0.987). The proposed ANN-GA model was also compared with models developed using particle swarm and random forest regressor.

Table 5 Comparative performance metrics

Model	On Native Test Set		On Common Valid Set	
	MAE	RMSE	MAE	RMSE
ANN-GA	3.065	3.883	4.221	6.032
Li et al. [26]	2.645	3.449	7.448	9.178
Al-Hashem et al. [32]	4.010	6.230	9.874	13.902
Amin et al. [81]	2.544	3.365	8.532	11.253
Saridemir et al. [82]	2.909	4.116	9.211	12.155
ANN-PSO	5.221	7.566	7.665	9.667
Random Forest	4.031	5.321	7.832	10.223

6.4.1 Performance Metrics of All Models Evaluated on their Common Data and Native Data Subsets

To facilitate an even-handed and non-biased evaluation of the models and their performance, the models were evaluated both on the native test subset associated with each model as well as on a subset of valid test samples that were shared by all models. This shared actual valid test sample provided the commonality among the data sets. A total of 246 data points were collected from across all data sources. By computing performance metrics on this common subset, discrepancies caused by differences in test data composition were eliminated, ensuring that all models were assessed using the exact same data points. These performance metrics are provided alongside the native test results in Table 5. Both values outline the changes in performance metrics. With respect to models having tested datasets of their own, the least errors were those of Amin et al. [81], with a mean absolute error (MAE) of 2.544 and a root mean square error (RMSE) of 3.365. The proposed ANN-GA model demonstrated comparable performance across the combined dataset encompassing all sources, highlighting its robustness. However, when restricting evaluation to the common valid data subset, all models experienced a decline in performance, as evidenced

by increases in MAE and RMSE values. Despite this overall reduction, the proposed ANN-GA model maintained the lowest errors, with an MAE of 4.221 and an RMSE of 6.032, outperforming other models on this unified dataset. These results emphasize that the ANN-GA model delivers consistent and desirable prediction accuracy regardless of whether the native test set or the common valid subset is considered. The repeated observation reinforces the model's reliability and effectiveness across varying evaluation

conditions, supporting its suitability for practical applications where consistent performance is critical.

6.4.2 Scatter Plot Analysis

Figure 14 compares the performance of the ANN-GA model and other existing equations. It shows the actual and predicted values of the compressive strength of RHA concrete using the 1192 datasets. The graphs contained multiple

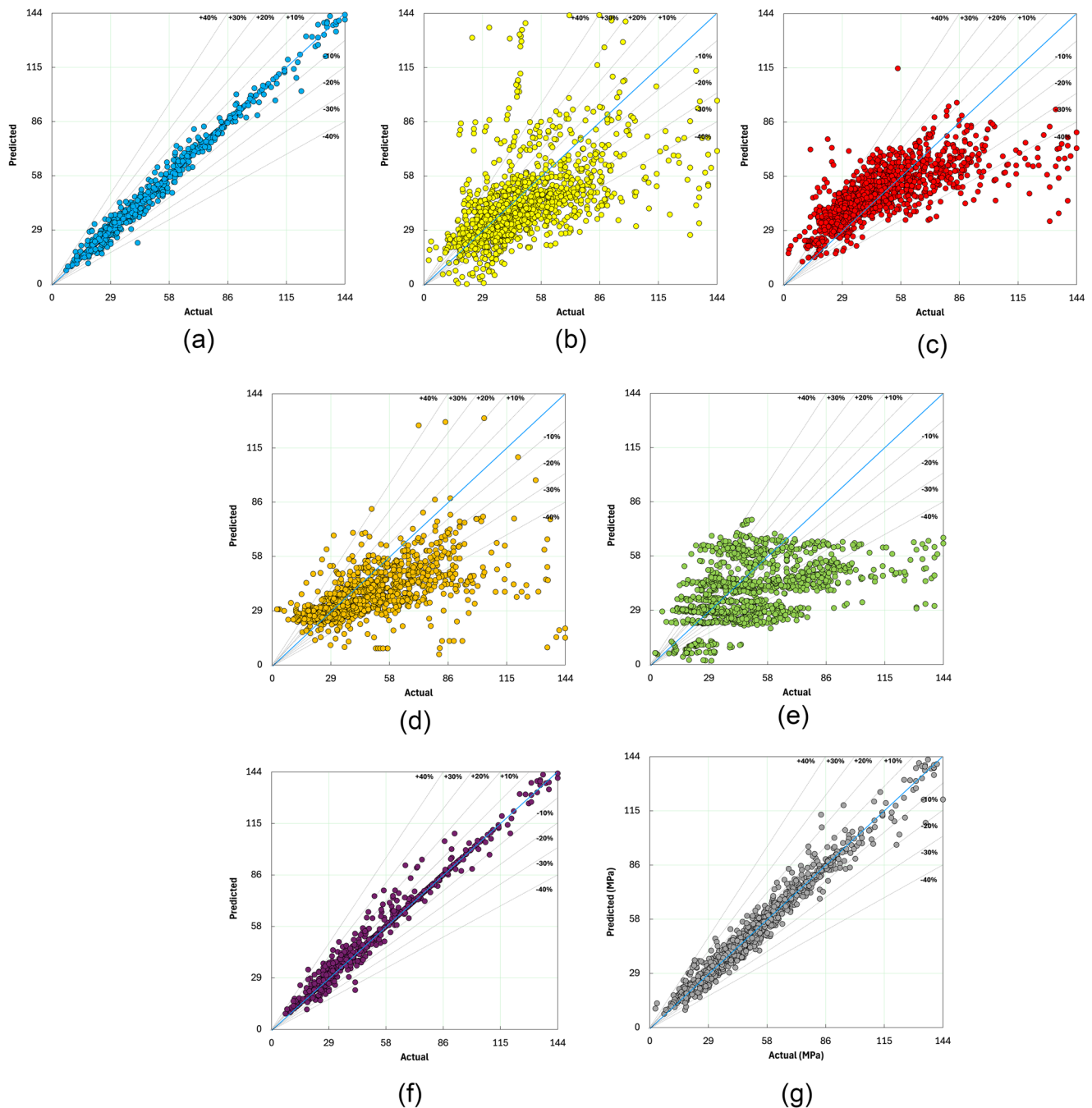


Fig. 14 Comparison of the scatter plot of the prediction performance of all models: (a) ANN-GA, (b) Li et al. [26], (c) Al-Hashem et al. [32], (d) Amin et al. [81], (e) saridemir et al. [82], (f) ANN-PSO, and (g) random forest regressor

diagonal lines, which includes a 45° line indicating a zero percent error, to indicate the accuracy of the models. The results reveal that the ANN-GA model outperforms the existing equations presented by Li et al. [26], Al-Hashem et al. [32], Amin et al. [81], and Saridemir et al. [82] in terms of the R-value. The lowest R-value came from Saridemir et al. [82] at 0.448, followed by Amin et al. [81] and Li et al. [26] with R-values of 0.526 and 0.538, respectively. In contrast, the lowest R-value among all the existing models was achieved by Al-Hashem et al. [32], with a value of 0.711. The two other machine learning models provided satisfactory performance demonstrating high performance indicators. The ANN-GA model exhibited the highest R-value among all the models at 0.988. Its graph illustrates a near linear arrangement near the 45° line, which reflects its high accuracy in predicting the values for the compressive strength of the RHA concrete.

6.4.3 Mean Absolute Percent Error Analysis

As shown in Fig. 15, bullseye diagrams are used to illustrate the performance of the ANN-GA model in comparison with other existing models presented by Li et al. [26], Al-Hashem et al. [32], Amin et al. [81], and Saridemir et al. [82]. Each diagram has five radial lines representing different thresholds of absolute error at constant interval of 20: specifically, 20, 40, 60, 80, and 100%. In comparing the performances of

the models, the mean average percentage error (MAPE) was utilized. The ANN-GA model achieved the lowest MAPE at 5.632%, indicating superior accuracy in prediction. This was followed by the model of Al-Hashem et al. [32] with a MAPE of 30.558%. Other models yielded higher errors: Li et al. [26] with 36.248%, Amin et al. [81] with 32.147%, and Saridemir et al. [82] with the highest MAPE at 37.506%. The two additional machine learning models also exhibited strong predictive performance. Specifically, the ANN-PSO achieved a MAPE of 12.224%, while the Random Forest Regression model demonstrated slightly better accuracy, yielding a lower MAPE of 9.952%. These results validate the robustness of the other machine learning approaches employed.

6.4.4 Taylor Diagram

The Taylor diagram offers a comprehensive graphical representation that simultaneously displays three key statistical measures: the standard deviation, the correlation coefficient (R), and the centered root mean square difference (RMSD) between predicted and actual values. In this analysis, the actual compressive strength serves as the reference field, while the test fields represent the predictive outputs of various models, namely the ANN-GA model, Li et al. [26], Al-Hashem et al. [32], Amin et al. [81], Saridemir et al. [82], ANN-PSO, and Random Forest Regressor. Out of the 179

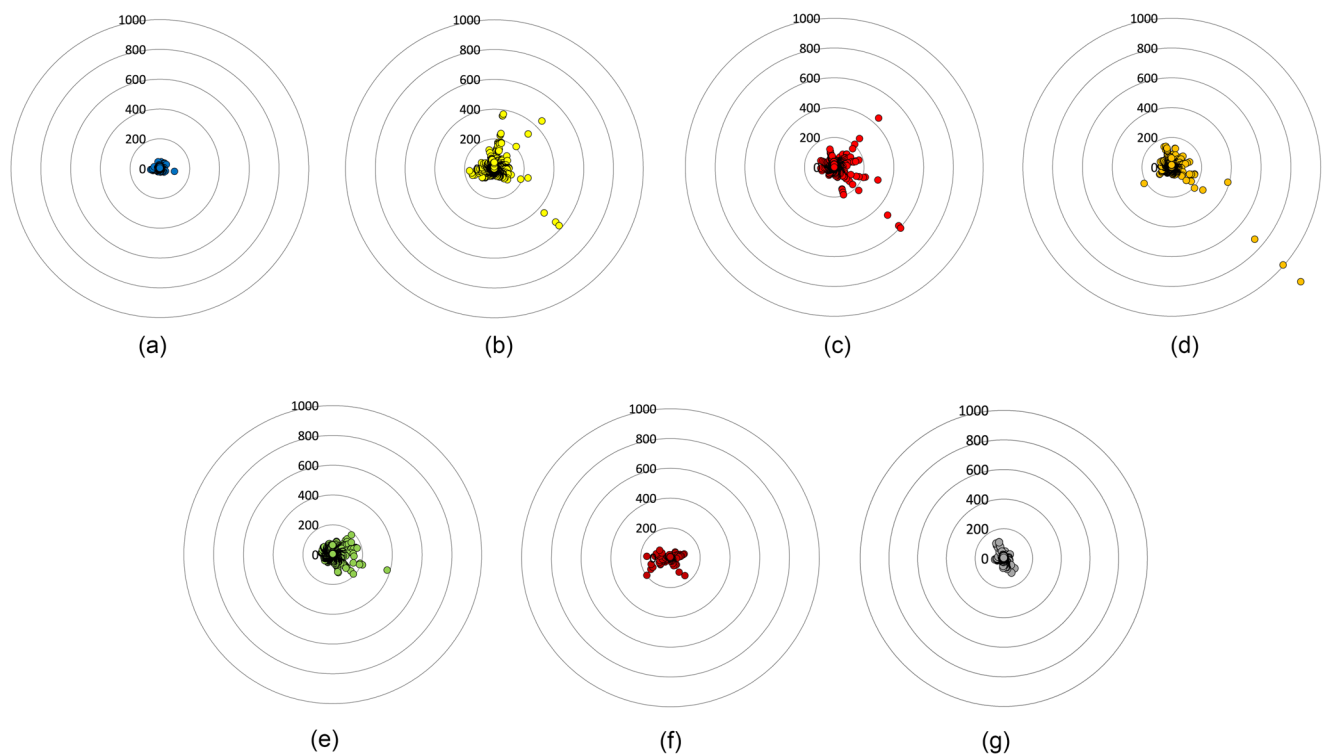
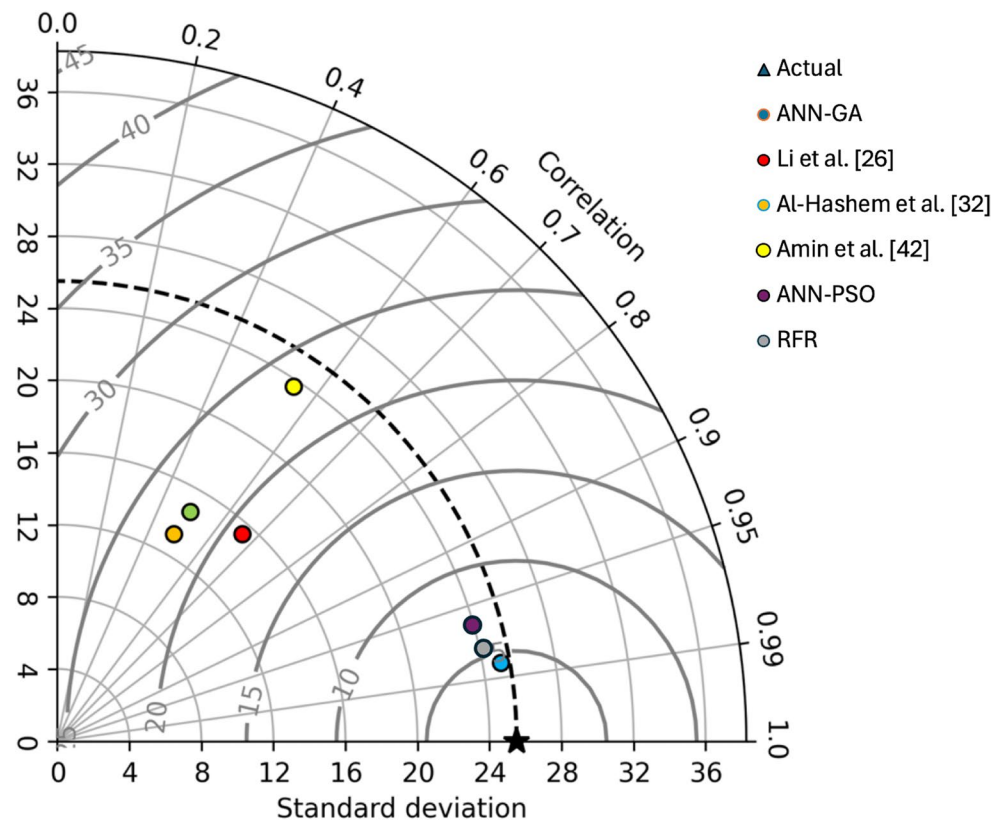


Fig. 15 Comparison of the absolute error analysis of the prediction performance of all models: (a) ANN-GA, (b) Li et al. [26] (c) Al-Hashem et al. [32], (d) Amin et al. [81], (e) saridemir et al. [82], (f) ANN-PSO, and (g) random forest regressor

Fig. 16 Comparison of the prediction performance of all models using a Taylor diagram



data points in the test subset, both the Li et al. [26] model and Saridemir et al. [82] model produced 179 valid predictions, yielding a VPR of 1.000. The Al-Hashem et al. [32] model followed closely with 178 valid predictions (VPR=0.99), while the Amin et al. [81] model returned 176 valid predictions (VPR=0.98). The Taylor diagram is shown in Fig. 16. The reference field, based on the actual compressive strength values, has a standard deviation of 25.32 and is plotted along the x-axis. Each model is then positioned according to its standard deviation (radial distance), R (angle from the x-axis), and centered RMSD (distance from the reference point), allowing a clear visual comparison of performance. As shown in the diagram, model performance based on standard deviation indicates that the ANN-GA Model has the closest radial distance to the reference, with a standard deviation of 24.32. This suggests that the model effectively replicates the variability observed in the actual compressive strength values.

A thorough evaluation of both the statistical metrics and the graphical representation in the Taylor diagram, the ANN-GA Model demonstrates clear superiority over all other existing models and other developed machine learning models. Although both ANN-PSO and RFR delivered satisfactory performance in predicting outcomes, the ANN-GA model consistently provided the best performance, yielding the most accurate and reliable results. Its superior predictive

Table 6 Comparative score analysis of all models based on different performance metrics

Models	R	R^2	MAE	RMSE	MAPE
ANN-GA	0.988	0.996	3.065	3.883	5.632
Li et al. [26]	0.539	0.291	19.342	23.782	27.667
Al-Hashem et al. [32]	0.711	0.506	15.334	17.920	17.231
Amin et al. [81]	0.526	0.276	8.221	10.377	13.337
Saridemir et al. [82]	0.447	0.200	20.118	24.736	29.853
ANN-PSO	0.968	0.937	5.221	7.566	9.218
Random Forest Regressor	0.973	0.947	4.031	5.321	7.432

capabilities were evident across various evaluation metrics, reinforcing its effectiveness in predicting the compressive strength of concrete. It consistently achieves the best performance across the three key indicators—standard deviation, R , and centered RMSD—and is plotted nearest to the reference point representing the actual compressive strength values. This proximity signifies that the ANN-GA Model not only captures the variability of the data but also maintains strong agreement in pattern and magnitude, making it the most accurate and reliable predictive model among those analyzed. While the other models showed promise, ANN-GA's optimization through genetic algorithms gave it a distinct advantage, making it the most robust and efficient approach among the compared methods.

6.4.5 Score Analysis

Score Analysis serves as an evaluation tool for the behavior and performance of the trained models. The ANN-GA model, mathematical models from Li et al. [26], Al-Hashem et al. [32], Amin et al. [81], Saridemir et al. [82], ANN-PSO, and Random Forest Regressor are compared based on their performance metrics such as Pearson correlation coefficient (R), mean absolute error (MAE), root mean squared error (RMSE), and percentage error (PE), as shown in Table 6.

The correlation coefficient (R) is a statistical measure that indicates the linear relationship between the dependent and independent quantitative variables. Its values range from -1 to 1 , with values closer to 1 signifying strong predictive accuracy. The Coefficient of Variation (R^2) the proportion of variance in the independent variable. Mean Absolute Error (MAE) is a commonly used metric to measure the average magnitude of errors in a set of predictions, without considering their direction. It gives a clear sense of how far predictions are from actual values on average. The Root Mean Square Error (RMSE) measures the average magnitude of prediction errors, while the Percentage Error (PE) quantifies the deviation between predicted and actual values as a percentage of the actual value. A lower MAE, RMSE, and MAPE indicate better model performance and accuracy. Among all the machine learning models evaluated in this study, the ANN-GA demonstrated the most outstanding predictive performance. It achieved the highest R value of 0.988 , indicating an excellent correlation between the predicted and actual compressive strength values of concrete. Furthermore, the ANN-GA model recorded the lowest MAE at 3.065 , the lowest RMSE at 3.883 , and the smallest MAPE of just 5.632% . These values collectively suggest that the ANN-GA model had the smallest average and squared deviations from the actual values, confirming its superior accuracy and consistency compared to the other models tested. Based on these key statistical indicators—high R and R^2 values and low MAE, RMSE, and MAPE—the ANN-GA model stands out as the most reliable and effective approach for predicting the compressive strength of concrete mixtures.

7 Three-Dimensional Parametric Analysis

This study considers seven parameters: Cement ($144\text{--}598\text{ kg/m}^3$), Water ($122.4\text{--}255\text{ kg/m}^3$), Fine Aggregate ($332\text{--}985\text{ kg/m}^3$), Coarse Aggregate ($625\text{--}1450\text{ kg/m}^3$), Rice Husk Ash (RHA) ($0\text{--}171\text{ kg/m}^3$), Superplasticizer ($0\text{--}24\text{ kg/m}^3$), and Concrete Age ($0\text{--}68$ days). To analyze the relationships between these parameters and their effect on the compressive strength of RHA concrete, three-dimensional pairwise

parametric models are developed. The six parameter pairs analyzed are: (1) water and cement, (2) water and RHA, (3) cement and RHA, (4) FA and CA, (5) age and SP, and (6) RHA and SP. The parametric analysis is conducted using the developed hybrid ANN-GA model, where each model visualizes the relationship between two independent variables (x and y axes) and the dependent variable, the predicted compressive strength (z axis). This is achieved by varying the values of the selected parameter pair while keeping the other parameters at their mean values. For each pair, 50 sets of data are generated and compiled into a 50×50 structured mesh table, enabling the creation of a three-dimensional surface plot parametric model. This model defines the specific relationship between the parameter pairs and their influence on compressive strength. The acceptable input ranges for the predictive model are defined strictly by the bounds of the training dataset. Inputs that fall outside these intervals are classified as out-of-distribution, meaning the model has not seen similar combinations during training. While the model can still generate predictions for such inputs, these outputs should be interpreted with caution, as their accuracy and reliability cannot be guaranteed. Additionally, to aid in practical engineering applications—such as structural design validation or quality assurance—prediction intervals have been established through statistical confidence analysis. For most test cases, a 95% confidence interval yields an estimated margin of error of approximately $\pm 1.4\text{ MPa}$.

7.1 Water-Cement vs Compressive Strength

The 3D parametric plot, illustrated in Fig. 17a, demonstrates the relationship between water content, cement content, and the compressive strength of RHA concrete. The parabolic shape of the surface reveals how different water–cement combinations influence strength. Peak compressive strength occurs at low water and high cement contents, consistent with the principle that a low w/c ratio reduces porosity, promotes efficient hydration, and results in a denser, stronger cement matrix. In contrast, the plot shows a pronounced decline in compressive strength when water content exceeds approximately 200 kg/m^3 , due to dilution of the cement paste and the formation of capillary voids that undermine mechanical integrity. The plot exhibits a threshold effect: increasing cement dosage improves the compressive strength only if it is accompanied by a controlled water dosage. For instance, as cement content increases to $450\text{--}650\text{ kg/m}^3$ and the water content is below 180 kg/m^3 , the compressive strength improves up to 140 MPa . Likewise, the upper-right zone of the plot reveals that compressive strength appears to plateau around 140 to 160 MPa . This observation indicates a saturation limit at high cement and moderate water content. Beyond these values, there will be no significant strength

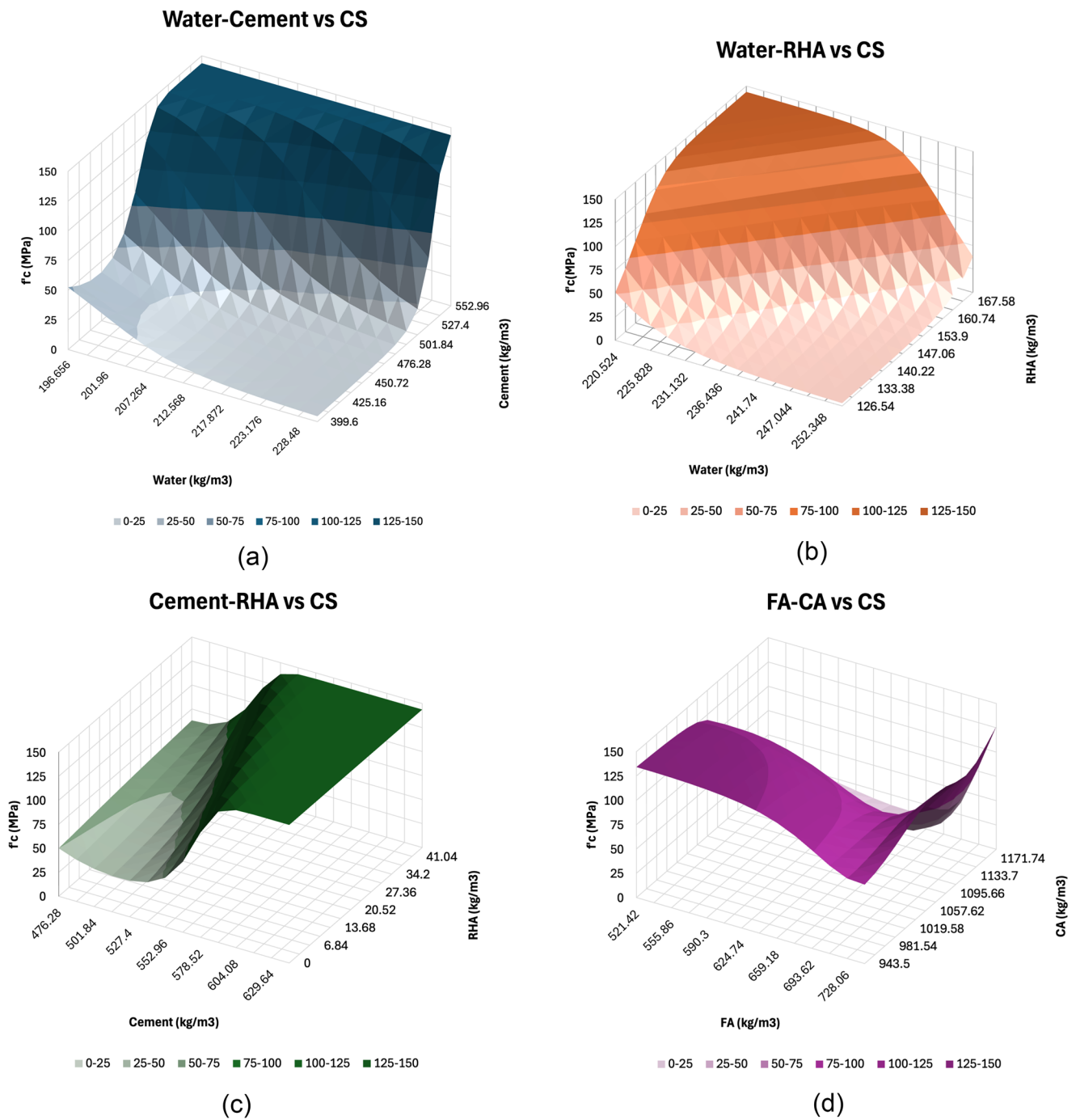


Fig. 17 3D parametric plot of parameter pairs vs compressive strength: (a) water-cement, (b) water-RHA, (c) cement-RHA, (d) FA-CA, (e) age-SP, and (f) RHA-SP

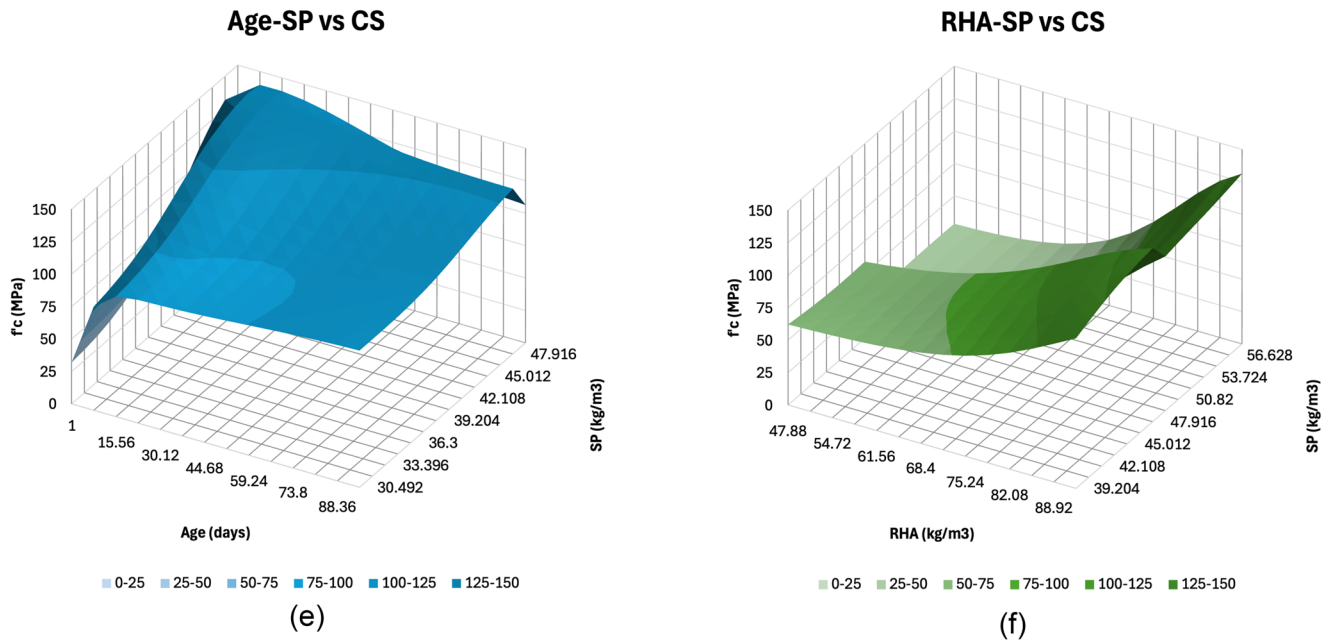


Fig. 17 (continued)

increase, unless admixtures are introduced. Piasta and Zarzycki [89] revealed that a decrease in w/c ratio causes decreased capillary porosity and enhanced aggregate-paste adhesion, which contributes to the development of concrete strength. Conversely, the concrete loses its strength as w/c increases resulting in larger pore sizes and increased water absorption. In support of this, a study by Abdullahi et al. [90] affirmed the abovementioned statement. According to their research, the compressive strength of RHA concrete is controlled by the water-cement ratio as much as the rice husk ash-water ratio. Their study specified an optimal w/c ratio of 0.50 to 0.55, rendering the most workable and durable mixes at or beyond 15% RHA replacement. Hence, it establishes the 3D parametric plot results that a balanced w/c ratio is crucial in maximizing strength.

7.2 Water-Rice Husk Ash (RHA) vs Compressive Strength

The 3D parametric plot for water and RHA, presented in Fig. 17b, demonstrates a pattern like that observed in the water-cement plot. As water content increases, the compressive strength of RHA concrete generally decreases, reflecting the typical weakening of the cementitious matrix caused by excess water. Conversely, increasing the RHA content initially enhances C compressive strength, with optimal performance observed within the range of 70–120 kg/m³. However, it is important to note that the influence of RHA on compressive strength is significantly less pronounced compared to that of cement. Beyond the optimal range,

additional RHA leads to a reduction in strength, as its role is limited to that of a supplementary cementitious material (SCM) rather than a complete replacement for cement. In line with this, several studies have established that the incorporation of RHA in concrete mixtures enhances compressive strength, primarily due to its pozzolanic properties. According to Hamidian et al. [91], these properties enable RHA to react with calcium hydroxide released during cement hydration, forming additional calcium silicate hydrate (C-S-H) gel, which contributes to the overall strength and durability of the concrete matrix. However, as noted by Vigneshwari et al. [92], this benefit is highly dependent on maintaining an optimal replacement level. Once this threshold is exceeded, the positive effects diminish, mainly due to the corresponding reduction in cement content, which remains the principal binder responsible for strength development. Further validating this inverse relationship, Fediuk et al. [93] emphasized the hygroscopic nature of RHA, noting its tendency to absorb moisture from both the environment and the mixed water. This increased water demand increases the w/c ratio, and excessive water, in turn, produces a more porous microstructure, reducing the density and mechanical integrity of the hardened concrete. Moreover, Abdullahi et al. [90] noted that increases in the RHA-to-water (r/w) ratio result in a significant decline in compressive strength, much more so compared to the c/w ratio. Thus, the interaction between RHA and water must be carefully balanced, as both excessive water and RHA content can undermine the compressive strength of concrete.

7.3 Cement-Rice Husk Ash (RHA) vs Compressive Strength

Figure 17c highlights the complex interaction between cement and rice husk ash in affecting concrete's compressive strength. As cement content increases, compressive strength rises sharply, indicating a strong positive correlation. In contrast, RHA exhibits a moderately increasing trend, where higher RHA levels combined with lower cement content result in enhanced strength. This behavior contrasts with the water-RHA interaction model, where high levels of both RHA and water tend to reduce compressive strength. The sustained increase in compressive strength with rising cement content is attributed to the direct formation of C-S-H, the principal binder responsible for strength in concrete. Furthermore, the SHAP analysis results from the study of Kashem et al. [6] revealed that cement's consistent contribution to strength ensures that, even as RHA content varies, high cement proportions continue to drive improvements in mechanical performance. On the other hand, RHAs influence on compressive strength is more indirect. At lower replacement levels, its high amorphous silica content promotes pozzolanic reactions with calcium hydroxide (CH), forming additional C-S-H and refining the microstructure. This contributes to improved strength and durability [32, 91]. However, this benefit peaks at an optimal level, beyond which excessive RHA reduces the cement content and available CH, weakening the matrix and lowering strength [93]. The figure clearly shows only the optimal range of RHA replacement, because beyond this level, a drop in compressive strength would be observed. Moreover, due to its porous and hygroscopic nature, excessive RHA increases water demand, which can further compromise concrete performance if not properly addressed. The relationship between RHA and cement isn't a direct proportion as it shows a decline in strength following RHAs optimal dosage. This of course indicates a tipping point beyond which further replacement becomes detrimental. Therefore, the model underscores that a moderate cement content paired with an optimal RHA replacement yields the highest compressive strength values, emphasizing the need for balanced mix design when incorporating supplementary cementitious materials like RHA.

7.4 Fine Aggregates (FA)-Coarse Aggregates (CA) vs Compressive Strength

The association among FA, CA, and the compressive strength of RHA concrete is shown in Fig. 17d. It demonstrates an intricate non-linear relationship between the three parameters. The combination of FA and CA results in the total aggregate volume of the concrete. The FA must be

increased if the CA decreases, and vice versa. Therefore, having an inverse relationship. In lower CA, the compressive strength decreases as FA increases while in higher CA, the compressive strength decreases as the FA increases which agrees with studies of Hamidian et al. [91]. This finding in the 3D model signifies that CA carries the increase in compressive strength of the concrete providing bulk and load-bearing capacity. When the proportion of CA is low, the concrete's strength relies on this limited amount. When there is an increase in FA when the CA is reduced, the total surface area of aggregates is also larger. Interfacial Transition Zone (ITZ) is the region where two components of a concrete meet. A large surface area makes the ITZ susceptible to weak points as the amount of cement paste bonding them reduces. The larger amount of FA, while making the spacing between the aggregates smaller and tends to separate the CA apart, results to ITZ percolation according to Lyu et al. [94] which contributes to the concrete's low compressive strength. The compressive strength varies as the FA increases, having a peak in certain proportion of FA in the mix and will increase again as the FA continues to rise. This non-linear relationship is evident in different CA amounts. The first increase in compressive strength indicates that the increase in FA improves the workability and packing density of the mix as the FA fills the voids between the CA particles leading to reduced air voids and improved density. These, together with optimized particle packing contribute to higher compressive strength. Beyond the optimal point, the compressive strength starts to decrease as the total surface area increases together with the amount of FA in the mix. This results in higher internal friction which makes it hard to achieve proper compaction which leads to more void spaces lowering the compressive strength of the concrete. The compressive strength increased again to high FA proportions as it may fill the void and promote higher packing density. However, higher FA greatly reduces the workability of the concrete needing more cement paste creating a less economical mix. There is an optimal number of FA to achieve a high compressive strength which aligns with hybrid ANN-GA. The 3D model demonstrates that improper ratio of the fine and CA could greatly impact the compressive strength of the RHA concrete.

7.5 Age-Superplasticizer (SP) vs Compressive Strength

The 3D parametric model for age and SP shown in Fig. 17e demonstrates a nonlinear relationship for both parameters. Compressive strength increases with age sharply within the first 14 days then increases moderately after, but its response to SP varies depending on dosage and curing time. Initially, as concrete ages, compressive strength increases rapidly due

to the hydration process, then slows as it enters the maturity phase. However, the influence of SP on strength is more complex, showing both enhancement and reduction effects depending on the dosage. Numerous studies have shown that the compressive strength of concrete increases as curing age increases. Vigneshwari et al. [92] shared that regardless of the curing method, whether normal or steam, the compressive strength of RHA concrete increases over time. In a study by Wagan et al. [95], RHA concrete tested at 1, 3, 7, and 28 days exhibited varying strength levels, with significant improvements observed as the curing age progressed. They further explained that this increase can be attributed to both the hydration process and microstructure development. As concrete cures, the hydration of cement continues, and the formation C-S-H progresses over time, resulting in increased strength as the concrete matures. With time, the microstructure of the concrete becomes denser and more interconnected. The ongoing hydration reactions fill voids and create a more compact structure, enhancing the overall strength of the concrete. As SP content increases (around 30–45 kg/m³), an optimal range is observed. Beyond this range, excessive SP leads to diminishing returns. The non-linearity of SP's effect on both curing age and compressive strength could be attributed to the fact that SPs are not a uniform material. According to Robeyst and Belie [96], PCE-based SPs generally cause a more pronounced retardation compared to other types, such as naphthalene-based SPs (PNS). Furthermore, Alya and Fatiha [97] added that at low dosage, SPs may slightly accelerate hydration due to improved dispersion and exposure of cement grains to water. However, at higher dosages, the retardation effect becomes dominant, leading to delayed setting times. These findings favorably confirm that SPs effect on age vary based on chemical structure, dosage efficiency, and interaction with cement hydration. In a study by Dvorkin et al. [98], different polymers used at their optimal concentrations—polycarboxylate ether, lignosulfonate, and naphthalene formaldehyde—exhibited compressive strengths of 60 MPa, 45 MPa, and 40 MPa for concrete mixtures based on low-clinker slag Portland cement after 28 days of curing. Likewise, Memon et al. [99] investigated the effects of curing type and duration, as well as SP addition (ranging from 0% to 2% by weight of cement), and found that an initial 7 to 14 days of wet curing followed by exposure to high ambient temperatures until the 28th day resulted in the best compressive strengths for concrete. These findings validate the observed trend, where moderate compatibility of the SP with the cement type, further explaining the observed variation. Reinforcing this pattern, Memon et al. [100] explained that SP dosages improve compressive strength by refining the microstructure through better particle packing, whereas overdosing can increase bleeding and entrapped air,

weakening the concrete. The interaction between age and SP further illustrates that while concrete generally strengthens over time, its long-term strength potential depends on SP dosage. While age typically follows an upward trend for compressive strength development, the model suggests that maintaining an optimal SP range, alongside adequate curing time, results in the highest compressive strength values.

7.6 Rice Husk Ash (RHA)-Superplasticizer (SP) vs Compressive Strength

Figure 17f shows the complex interaction between the RHA and SP and their influence on compressive strength. As the RHA content increases, the compressive strength follows a generally nonlinear upward trend. In contrast, the effect of SP shows a different pattern: initially, increasing SP enhances the compressive strength of RHA concrete, but beyond a certain point, further addition of SP leads to a decline in strength. In a study conducted by Kashem et al. [6], a direct proportional relationship was observed between the RHA content and the compressive strength of RHA concrete. They found out that increasing the RHA concentration led to a corresponding increase in compressive strength. Specifically, when their RHA concentration increased from 25 kg/m³ to 170 kg/m³, the corresponding compressive strength also increased from 42 MPa to 70 MPa due to the finer particle size and high pozzolanic activity of RHA. These findings confirm the observed trend, where the compressive strength exhibited a general increase as the RHA content was also increased. Moreover, the gradual and non-linear increase of compressive strength is due to the RHAs pozzolanic activity and material variability. Compared to cement particles, RHA typically has a finer particle size. This finer size allows RHA to effectively fill the voids between cement particles and within the cement paste matrix, leading to a denser and more compact microstructure [101]. In another study conducted by Fode et al. [102], they have observed the effect of SP on concrete where they found out that SP initially improves the compressive strength of concrete due to the breaking up of water trapped within cement flocs which increases the concrete's density and reduces voids. But as more SP is added beyond its optimal amount, the strength turns to a downward and horizontal trend due to excessive fluidity, segregation, and bleeding. This damages the cohesiveness of concrete as it increases air content in the cement-to-water interface which lessens the durability of concrete [103]. These findings are consistent with the performance of the SP exhibited in Fig. 17(f).

These findings highlight the overall performance behavior of RHA concrete in relation to its mixed components. The parametric analysis revealed that compressive strength is strongly influenced by the balance among the parameters:

Compressive Strength of RHA Concrete

Components		
Water:	180	(122.4-255) kg/m ³
Cement:	360	(144-598) kg/m ³
Fine Aggregates:	750	(332-985) kg/m ³
Coarse Aggregates:	840	(625-1450) kg/m ³
Rice Husk Ash:	40	(0-171) kg/m ³
Age:	28	(0-68) days
Superplasticizer:	15	(0-24) kg/m ³

Compressive Strength

86.27

MPa

Fig. 18 Graphical user interphase to predict the compressive strength of RHA concrete

water, cement, FA, CA, RHA, age, and SP. While RHA contributes positively at moderate levels, its effectiveness depends on proper proportioning with other key ingredients. The trends observed in this study align with previous experimental results, reinforcing the potential of RHA concrete as a sustainable and high-performing material when mixed and used appropriately. Figure 18 illustrates the graphical user interface (GUI) developed for predicting the compressive strength of concrete mixtures. This application was created using the Visual Basic programming language, which offers a user-friendly and interactive platform. The GUI allows users to input the various components of a concrete mixture into clearly labeled textboxes. Once all required values are entered, the program instantly calculates the predicted compressive strength based on the input parameters. This immediate feedback enhances usability and efficiency, making it particularly useful for field applications. The tool is especially beneficial for practicing engineers, as it serves as a quick and nondestructive means of estimating the compressive strength of concrete mixtures that incorporate rice husk ash (RHA) as a partial cement replacement. With its intuitive interface and rapid calculation capability, the GUI has the potential to assist in optimizing mix designs while promoting sustainable construction practices. The proposed GUI is designed for practical engineering applications, particularly in the prediction of concrete compressive

strength for mix design and quality assurance (QA) purposes. The valid input ranges for the model are based on the training data and are as follows: Cement: 144–598 kg/m³, Water: 122.4–255 kg/m³, Fine Aggregate: 332–985 kg/m³, Coarse Aggregate: 625–1450 kg/m³, Rice Husk Ash (RHA): 0–171 kg/m³, Superplasticizer: 0–24 kg/m³, and Age: 0–68 days. Inputs beyond these ranges are considered out-of-distribution. The model may still produce outputs, but accuracy and reliability are not guaranteed. For such cases, a warning mechanism is incorporated at the software or interface level to flag extrapolated predictions. Practical prediction intervals have also been computed using a confidence interval analysis to support engineering design and QA decisions. For most predictions, a 95% confidence interval of ± 1.4 MPa margin of error can be expected.

The prediction model developed for estimating the compressive strength of rice husk ash (RHA) concrete introduces a novel and valuable non-destructive approach within the field of sustainable construction materials. This model offers engineers and researchers an effective means to assess the performance of eco-friendly concrete without the need for extensive physical testing, thereby saving both time and resources. Furthermore, the user-friendly GUI developed alongside the model enhances its practical applicability. It is cost-effective and designed to support real-world decision-making in the design and optimization of green concrete

mixtures. By accurately predicting the strength of RHA concrete, the model will contribute to the reduction of excessive cement use, which not only lowers material costs but also plays a significant role in reducing CO₂ emissions associated with cement production. As the construction industry moves toward more sustainable practices, this tool provides an important step forward in promoting environmentally responsible concrete technologies.

8 Limitations

While this study offers meaningful insights into predicting the compressive strength of RHA concrete using a hybrid ANN-GA model, certain limitations should be recognized. The dataset primarily consists of data corresponding to early-age concrete, which may limit the model's accuracy in forecasting long-term strength performance. Though early strength is used in the design of concrete structures, the long-term behavior of concrete with RHA is also essential. The input data set must fall within the indicated range of each concrete mix component, as inputs that fall outside the trained data range may reduce prediction reliability. It is important to emphasize that the predictive capability of the developed model is specifically tailored to concrete mixtures incorporating rice husk ash (RHA) as a supplementary cementitious material. As such, the model is not expected to yield accurate or reliable results when applied to other types of supplementary cementitious materials, due to differences in chemical and physical properties. Moreover, the scope of the study is confined to predicting compressive strength, without addressing other important durability-related properties such as permeability, chloride ion penetration, and shrinkage. These parameters are critical for assessing the long-term structural performance and service life of concrete. As such, the absence of these factors represents a limitation in the model's comprehensive applicability. These constraints highlight areas for future research, particularly in expanding the dataset to include a broader age range and incorporating additional performance indicators to enhance the model's robustness and practical relevance.

9 Conclusions

The objective of this study is to predict the compressive strength of RHA concrete using a hybrid neuro-genetic algorithm (ANN-GA). The model's predictive capability is

assessed based on seven key input parameters: water (kg/m³), cement (kg/m³), FA (kg/m³), CA (kg/m³), RHA (kg/m³), age (days), and SP (kg/m³). The performance of the hybrid ANN-GA model is evaluated through different statistical metrics, model comparison, and parametric analysis. Results demonstrated that the proposed hybrid ANN-GA model outperformed other existing approaches in terms of accuracy and reliability when applied to the consolidated and analyzed dataset. Based on these findings, the study concludes with the following observations:

1. The neural network combined with a genetic algorithm exhibited robust predictive power in modeling the compressive strength of RHA concrete. By harnessing the neural network's strength in capturing complex nonlinear relationships and the genetic algorithm's proficiency in optimizing model parameters, the hybrid approach efficiently identified the best network configuration. This synergy enabled the model to achieve enhanced accuracy and reliability in its predictions.
2. The model developed using an artificial neural network optimized by a genetic algorithm, featuring a single hidden layer with 36 neurons, demonstrated excellent performance in predicting the compressive strength of RHA concrete. It achieved a high overall correlation coefficient (R-value) of 0.988, mean absolute error (MAE) of 3.065, and mean squared error (RMSE) of 3.883, reflecting a strong fit to the data. Overall, the hybrid ANN-GA model delivered predictions that closely matched the observed values, confirming its robustness and accuracy.
3. The amount of cement and aggregates are found to be the most influential parameters in the compressive strength of RHA concrete. The combined effect of these components governs the microstructure and mechanical integrity of the concrete, underscoring their importance in optimizing mix design for enhanced compressive strength.
4. Among the various compressive strength models reviewed in the literature, the proposed ANN-GA 7-36-1 model demonstrated superior performance across a broad spectrum of evaluation metrics. These models offer a promising nondestructive alternative for assessing the compressive strength of RHA concrete, combining accuracy with practical applicability in quality control [104].

Appendix

ANN-GA 7-36-1 model architecture and permutation importance

Hidden Neurons	Input Layer							Output Layer	Hidden Layer Biases
	Water	Cement	FA	CA	RHA	Age	SP	CS	
1	-1.202	-0.668	-0.222	-2.369	0.267	-0.887	0.168	-0.486	-0.542
2	-2.080	0.689	-0.165	0.269	3.692	-0.556	0.283	0.385	0.634
3	-2.203	-1.654	0.298	-0.012	-2.841	-2.188	-0.518	-3.719	-1.290
4	1.057	-3.810	1.526	0.083	-1.891	-2.077	-5.149	1.214	-1.921
5	-1.477	0.267	-0.887	0.363	-1.411	1.988	0.042	-3.614	-0.725
6	-1.864	3.692	-0.556	1.127	-2.678	4.553	-0.179	0.717	1.077
7	-5.030	-2.841	-2.188	0.479	-0.447	-1.928	3.778	-3.570	1.583
8	3.825	-0.912	-0.314	0.096	-0.530	2.046	0.414	0.860	-2.784
9	3.256	2.230	2.178	-0.222	-0.085	-4.428	0.078	2.547	3.537
10	-3.570	1.583	-1.370	2.226	-1.831	-5.383	-1.489	-0.649	-1.284
11	0.860	-2.784	5.260	3.851	-0.029	-1.244	-1.370	-0.668	-0.222
12	2.547	3.537	-0.869	4.202	-1.615	-5.128	5.260	0.689	-0.165
13	-0.649	-1.284	-1.871	-2.497	-2.878	4.716	-0.869	-1.654	0.298
14	-0.668	-0.222	-2.369	-0.926	1.707	0.493	-1.871	-3.810	1.526
15	0.689	-0.165	0.269	3.857	-2.027	3.392	-2.369	0.267	-0.887
16	-1.654	0.298	-0.012	1.929	0.731	1.863	0.269	3.692	-0.556
17	-3.810	1.526	0.083	1.065	0.268	3.486	-0.012	-2.841	-2.188
18	2.226	-1.831	-5.383	-1.489	-0.649	-1.284	0.083	-1.891	-3.810
19	3.851	-0.029	-1.244	-1.370	-0.668	-0.222	0.363	-1.411	0.267
20	4.202	-1.615	-5.128	5.260	0.689	-0.165	1.127	-2.678	3.692
21	-2.497	-2.878	4.716	-0.869	-1.654	0.298	0.479	-0.447	-2.841
22	-0.926	1.707	0.493	-1.871	-3.810	1.526	0.096	-0.530	-1.891
23	3.857	-2.027	3.392	-2.369	0.267	-0.887	-0.222	-0.085	-1.411
24	1.929	0.731	1.863	0.269	3.692	-0.556	2.226	-1.831	-2.678
25	1.065	0.268	3.486	-0.012	-2.841	-2.188	3.851	-0.029	-3.810
26	4.200	0.304	2.110	0.083	-1.891	-2.077	4.202	-1.615	0.267
27	-0.486	0.240	-3.364	0.363	-1.411	1.988	-2.497	-2.878	3.692
28	0.385	0.419	-0.549	1.127	-2.678	4.553	-0.926	1.707	-2.841
29	-3.719	0.199	3.380	0.479	-0.447	-1.928	3.857	-2.027	-1.891
30	1.214	5.736	-1.367	0.096	-0.530	2.046	1.929	0.731	-1.411
31	5.322	2.230	2.178	-0.222	-0.085	-4.428	0.689	0.268	-2.678
32	-3.570	1.583	-1.370	2.226	-1.831	-5.383	-0.165	0.304	-0.447
33	0.860	-2.784	5.260	3.851	-0.029	-1.244	0.269	1.240	-0.530
34	2.547	3.537	-0.869	4.202	-1.615	-5.128	3.857	0.419	-0.085
35	-0.649	-1.284	-1.871	-2.497	-2.878	4.716	-2.027	0.199	-1.831
36	-0.668	-0.222	-2.369	-0.926	1.707	0.493	3.392	2.736	-0.029
PI%	12.50%	16.88%	15.75%	15.44%	12.96%	11.53%	14.94%	Output Biases	0.453
Rank	6	1	2	3	5	7	4		

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Declarations

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